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Enhanced hybrid LIDAR with high-speed scanning

Abstract

An optical arrangement for a LIDAR system includes a laser, scanner, fold mirror, lens, and sensors. Adding a wedge-shaped optical element creates an auxiliary field of view, when positioned to overlie an edge portion of the lens. The wedge-shaped optical element may have a variable transparency preventing reception by the sensors of excessive signals from close targets. The sensors may be: a plurality of sensors with unique lengths/widths, arranged to form a linear array; or sensors with the same length/width, but positioned to form an arcuate shape, compensating for an arcuate scan line; or two columns of sensors, one with a field of view in focus at far field positions, and the other in focus at near field positions, correcting for parallax effect/error; or a column of highly sensitive sensors surrounded by a plurality of guard rail sensors having higher immunity to damage from stray laser light.

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9366938	12/2015	Anderson	N/A	N/A
9369689	12/2015	Tran	N/A	N/A
9378463	12/2015	Zeng	N/A	N/A
9383447	12/2015	Schmitt	N/A	N/A
9383753	12/2015	Tempelton	N/A	N/A
9407285	12/2015	Kozak	N/A	N/A
9420177	12/2015	Pettegrew	N/A	N/A
9420264	12/2015	Gilliland	N/A	N/A
9425654	12/2015	Lenius	N/A	N/A
9448110	12/2015	Wong	N/A	N/A
9453907	12/2015	Zheleznyak	N/A	N/A
9453914	12/2015	Stettner	N/A	N/A
9453941	12/2015	Stainvas Olshansky	N/A	N/A
9465112	12/2015	Stettner	N/A	N/A
9470520	12/2015	Schwarz	N/A	N/A
9476968	12/2015	Anderson	N/A	N/A
9476983	12/2015	Zeng	N/A	N/A
9489746	12/2015	Sebastian	N/A	N/A
9495466	12/2015	Geringer	N/A	N/A
9519979	12/2015	Hilde	N/A	N/A
9523772	12/2015	Rogan	N/A	N/A
9525863	12/2015	Nawasra	N/A	N/A

9529087	12/2015	Stainvas Olshansky	N/A	N/A
9530062	12/2015	Nguyen	N/A	N/A
9547074	12/2016	Schulz	N/A	N/A
9575162	12/2016	Owechko	N/A	N/A
9575164	12/2016	Kim	N/A	N/A
9575184	12/2016	Gilliand	N/A	N/A
9575341	12/2016	Heck	N/A	N/A
9588220	12/2016	Rondeau	N/A	N/A
9599468	12/2016	Walsh	N/A	N/A
9599714	12/2016	Imaki	N/A	N/A
9602224	12/2016	McLaughlin	N/A	N/A
9606236	12/2016	Rojas	N/A	N/A
9625580	12/2016	Kotelnikov	N/A	N/A
9625582	12/2016	Gruver	N/A	N/A
9651658	12/2016	Pennecot	N/A	N/A
9658322	12/2016	Lewis	N/A	N/A
9658337	12/2016	Ray	N/A	N/A
9678199	12/2016	Hutson	N/A	N/A
9702975	12/2016	Brinkmeyer	N/A	N/A
9710714	12/2016	Chen	N/A	N/A
9735885	12/2016	Sayyah	N/A	N/A
9753124	12/2016	Hayes	N/A	N/A
9753462	12/2016	Gilliland	N/A	N/A
9759809	12/2016	Derenick	N/A	N/A
9772399	12/2016	Schwarz	N/A	N/A
9778362	12/2016	Rondeau	N/A	N/A
9784840	12/2016	Pedersen	N/A	N/A
9790924	12/2016	Bayon	N/A	N/A
9791555	12/2016	Zhu	N/A	N/A
9791557	12/2016	Wyrwas	N/A	N/A
9797995	12/2016	Gilliland	N/A	N/A
9804264	12/2016	Villeneuve	N/A	N/A
9810775	12/2016	Welford	N/A	N/A
9810776	12/2016	Sapir	N/A	N/A
9810777	12/2016	Williams	N/A	N/A
9810786	12/2016	Welford	N/A	N/A
9812838	12/2016	Villeneuve	N/A	N/A
9823118	12/2016	Doylend	N/A	N/A
9823350	12/2016	Fluckiger	N/A	N/A
9823351	12/2016	Haslim	N/A	N/A
9823353	12/2016	Eichenholz	N/A	N/A
9830509	12/2016	Zang	N/A	N/A
9831630	12/2016	Lipson	N/A	N/A
9834209	12/2016	Stettner	N/A	N/A
9841495	12/2016	Campbell	N/A	N/A
9851433	12/2016	Sebastian	N/A	N/A
9851442	12/2016	Lo	N/A	N/A
RE46672	12/2017	Hall	N/A	N/A
9857473	12/2017	Kim	N/A	N/A
9860770	12/2017	McLaughlin	N/A	N/A

9869753	12/2017	Eldada	N/A	N/A
9869754	12/2017	Campbell	N/A	N/A
9870512	12/2017	Rogan	N/A	N/A
9872010	12/2017	Tran	N/A	N/A
9874635	12/2017	Eichenholz	N/A	N/A
9877009	12/2017	Tran	N/A	N/A
9880263	12/2017	Droz	N/A	N/A
9880281	12/2017	Gilliland	N/A	N/A
9881220	12/2017	Korvadi	N/A	N/A
9882433	12/2017	Lenius	N/A	N/A
9885778	12/2017	Dussan	N/A	N/A
9891711	12/2017	Lee	N/A	N/A
9892567	12/2017	Binion	N/A	N/A
9897687	12/2017	Campbell	N/A	N/A
9897689	12/2017	Dussan	N/A	N/A
9904375	12/2017	Donnelly	N/A	N/A
9905032	12/2017	Rogan	N/A	N/A
9905987	12/2017	Seo	N/A	N/A
9905992	12/2017	Welford	N/A	N/A
9910136	12/2017	Heo	N/A	N/A
9910139	12/2017	Pennecot	N/A	N/A
9910155	12/2017	Lundquist	N/A	N/A
9915726	12/2017	Bailey	N/A	N/A
9921297	12/2017	Jungwirth	N/A	N/A
9921307	12/2017	Schmalengurg	N/A	N/A
9927524	12/2017	Kaiser	N/A	N/A
9933513	12/2017	Dussan	N/A	N/A
9933514	12/2017	Gylys	N/A	N/A
9945950	12/2017	Newman	N/A	N/A
9958545	12/2017	Eichenholz	N/A	N/A
9971024	12/2017	Schwarz	N/A	N/A
9971035	12/2017	Imaki	N/A	N/A
9983297	12/2017	Hall	N/A	N/A
9983590	12/2017	Templeton	N/A	N/A
9985071	12/2017	Irish	N/A	N/A
9989969	12/2017	Eustice	N/A	N/A
10000000	12/2017	Marron	N/A	N/A
10012474	12/2017	Teetzel	N/A	N/A
10012723	12/2017	Lindskog	N/A	N/A
10012732	12/2017	Eichenholz	N/A	N/A
10018711	12/2017	Sebastian	N/A	N/A
10018725	12/2017	Liu	N/A	N/A
10018726	12/2017	Hall	N/A	N/A
10019803	12/2017	Venable	N/A	N/A
10024964	12/2017	Pierce	N/A	N/A
10031214	12/2017	Rosenweig	N/A	N/A
10031231	12/2017	Zermas	N/A	N/A
10031232	12/2017	Zohar	N/A	N/A
10032369	12/2017	Korvadi	N/A	N/A
10036801	12/2017	Reterrath	N/A	N/A

10036803	12/2017	Pacala	N/A	N/A
D826746	12/2017	Qiu	N/A	N/A
10042042	12/2017	Miremadi	N/A	N/A
10042043	12/2017	Dussan	N/A	N/A
10042159	12/2017	Dussan	N/A	N/A
10046187	12/2017	Doten	N/A	N/A
10048359	12/2017	Zhelenyzak	N/A	N/A
10048374	12/2017	Hall	N/A	N/A
10054841	12/2017	Nomura	N/A	N/A
10061019	12/2017	Campbell	N/A	N/A
10061020	12/2017	Slobodyanyuk	N/A	N/A
10061266	12/2017	Christmas	N/A	N/A
10067230	12/2017	Smits	N/A	N/A
10073166	12/2017	Dussan	N/A	N/A
10078133	12/2017	Dussan	N/A	N/A
10078137	12/2017	Ludwig	N/A	N/A
10088557	12/2017	Yeun	N/A	N/A
10088558	12/2017	Dussan	N/A	N/A
10094657	12/2017	Kiss	N/A	N/A
10094916	12/2017	Droz	N/A	N/A
10094925	12/2017	LaChapelle	N/A	N/A
10094928	12/2017	Josset	N/A	N/A
10107914	12/2017	Kalscheur	N/A	N/A
10107915	12/2017	Rozenzweig	N/A	N/A
10109208	12/2017	Cherepinsky	N/A	N/A
10114109	12/2017	Gazit	N/A	N/A
10114112	12/2017	Slobodyanyuk	N/A	N/A
10115024	12/2017	Stein	N/A	N/A
10120076	12/2017	Scheim	N/A	N/A
10121813	12/2017	Eichenholz	N/A	N/A
10126411	12/2017	Gilliland	N/A	N/A
10126412	12/2017	Eldada	N/A	N/A
10131446	12/2017	Stambler	N/A	N/A
10132928	12/2017	Eldada	N/A	N/A
10139478	12/2017	Gaalema	N/A	N/A
10142538	12/2017	Hurd	N/A	N/A
10145941	12/2017	Lee	N/A	N/A
10145944	12/2017	Shchemelinin	N/A	N/A
10145945	12/2017	Harada	N/A	N/A
10148060	12/2017	Hong	N/A	N/A
10151836	12/2017	O'Keeffe	N/A	N/A
10168423	12/2018	Lombrozo	N/A	N/A
10168429	12/2018	Maleki	N/A	N/A
10175344	12/2018	Jungwirth	N/A	N/A
10175361	12/2018	Haines	N/A	N/A
10180493	12/2018	Eldada	N/A	N/A
10185027	12/2018	O'Keeffe	N/A	N/A
10185028	12/2018	Dussan	N/A	N/A
10185033	12/2018	Justice	N/A	N/A
10191156	12/2018	Steinberg	N/A	N/A

10197669	12/2018	Hall	N/A	N/A
10197676	12/2018	Slobodyyanyuk	N/A	N/A
10197765	12/2018	Schulz	N/A	N/A
10203399	12/2018	Retterath	N/A	N/A
10203401	12/2018	Sebastian	N/A	N/A
10209349	12/2018	Dussan	N/A	N/A
10209359	12/2018	Russell	N/A	N/A
10209709	12/2018	Peters	N/A	N/A
10214299	12/2018	Jackowski	N/A	N/A
10215846	12/2018	Carothers	N/A	N/A
10215847	12/2018	Scheim	N/A	N/A
10215848	12/2018	Dussan	N/A	N/A
10215859	12/2018	Steinberg	N/A	N/A
10222474	12/2018	Raring	N/A	N/A
10222477	12/2018	Keilaf	N/A	N/A
10223806	12/2018	Luo	N/A	N/A
10223807	12/2018	Luo	N/A	N/A
10241196	12/2018	Bailey	N/A	N/A
10241198	12/2018	LaChappelle	N/A	N/A
10247811	12/2018	Clifton	N/A	N/A
10254402	12/2018	Lane	N/A	N/A
10254405	12/2018	Campbell	N/A	N/A
10261006	12/2018	Ray	N/A	N/A
10261187	12/2018	Halmos	N/A	N/A
10262234	12/2018	Li	N/A	N/A
10267898	12/2018	Campbell	N/A	N/A
10267918	12/2018	LaChapelle	N/A	N/A
10274377	12/2018	Rabb	N/A	N/A
10274599	12/2018	Schmalenberg	N/A	N/A
D849573	12/2018	Haban	N/A	N/A
10281254	12/2018	Ginsberg	N/A	N/A
10281322	12/2018	Doyland	N/A	N/A
10281564	12/2018	Low	N/A	N/A
10281581	12/2018	Lipson	N/A	N/A
10281582	12/2018	Elooz	N/A	N/A
10288736	12/2018	Lipson	N/A	N/A
10288737	12/2018	Mooney	N/A	N/A
10295656	12/2018	Li	N/A	N/A
10295660	12/2018	McMichael	N/A	N/A
10295668	12/2018	LaChapelle	N/A	N/A
10295670	12/2018	Stettner	N/A	N/A
10295671	12/2018	Grazit	N/A	N/A
10295672	12/2018	Abari	N/A	N/A
10295673	12/2018	Tucker	N/A	N/A
10302746	12/2018	O'Keeffee	N/A	N/A
10302749	12/2018	Droz	N/A	N/A
D850306	12/2018	Bainter	N/A	N/A
10310058	12/2018	Campbell	N/A	N/A
10310087	12/2018	Laddha	N/A	N/A
10317529	12/2018	Shy	N/A	N/A

10317533	12/2018	Cherepinsky	N/A	N/A
10324170	12/2018	Enberg	N/A	N/A
10324185	12/2018	McWhirter	N/A	N/A
10330777	12/2018	Popovich	N/A	N/A
10330778	12/2018	Kaneda	N/A	N/A
10330780	12/2018	Hall	N/A	N/A
10331956	12/2018	Solar	N/A	N/A
10337996	12/2018	Blagojevic	N/A	N/A
10338201	12/2018	Slobodyyanyuk	N/A	N/A
10338202	12/2018	Mashtare	N/A	N/A
10338220	12/2018	Raring	N/A	N/A
10338224	12/2018	Eken	N/A	N/A
10338225	12/2018	Boehmke	N/A	N/A
10340651	12/2018	Drummer	N/A	N/A
10345446	12/2018	Raring	N/A	N/A
10345447	12/2018	Hicks	N/A	N/A
10346695	12/2018	Clifford	N/A	N/A
10351103	12/2018	Yeo	N/A	N/A
10353057	12/2018	Suzuki	N/A	N/A
10353074	12/2018	Justice	N/A	N/A
10353075	12/2018	Buskila	N/A	N/A
10359507	12/2018	Berger	N/A	N/A
10366282	12/2018	Lee	N/A	N/A
10372138	12/2018	Gilliland	N/A	N/A
10377373	12/2018	Stettner	N/A	N/A
10379135	12/2018	Maryfield	N/A	N/A
10379205	12/2018	Dussan	N/A	N/A
10379220	12/2018	Smits	N/A	N/A
10379540	12/2018	Droz	N/A	N/A
10386464	12/2018	Dussan	N/A	N/A
10386465	12/2018	Hall	N/A	N/A
10386467	12/2018	Dussan	N/A	N/A
10386487	12/2018	Wilton	N/A	N/A
10386488	12/2018	Ridderbusch	N/A	N/A
10393863	12/2018	Sun	N/A	N/A
10393877	12/2018	Hall	N/A	N/A
10394345	12/2018	Donnelly	N/A	N/A
10401480	12/2018	Gaalema	N/A	N/A
10401484	12/2018	Lee	N/A	N/A
10401500	12/2018	Yang	N/A	N/A
10401866	12/2018	Rust	N/A	N/A
10408926	12/2018	Slobodyyabnyuk	N/A	N/A
10408936	12/2018	Van Voorst	N/A	N/A
10408939	12/2018	Kim	N/A	N/A
10408940	12/2018	O'Keeffe	N/A	N/A
10416292	12/2018	de Mersseman	N/A	N/A
10418776	12/2018	Welford	N/A	N/A
10422862	12/2018	Gnecchi	N/A	N/A
10422863	12/2018	Choi	N/A	N/A
10422865	12/2018	Irish	N/A	N/A

10429243	12/2018	Yu	N/A	N/A
10429495	12/2018	Wang	N/A	N/A
10429496	12/2018	Weinberg	N/A	N/A
10429507	12/2018	Sebastian	N/A	N/A
10429511	12/2018	Bosetti	N/A	N/A
10430970	12/2018	Bier	N/A	N/A
10436882	12/2018	Meng	N/A	N/A
10436904	12/2018	Moss	N/A	N/A
10436907	12/2018	Murray	N/A	N/A
10444330	12/2018	Stann	N/A	N/A
10444356	12/2018	Wu	N/A	N/A
10444362	12/2018	Schaefer	N/A	N/A
10444367	12/2018	Lodden	N/A	N/A
10445928	12/2018	Nehmadi	N/A	N/A
10447973	12/2018	Droz	N/A	N/A
10451716	12/2018	Hughes	N/A	N/A
10451740	12/2018	Pei	N/A	N/A
10451742	12/2018	Christmas	N/A	N/A
10458904	12/2018	Batholomew	N/A	N/A
10466342	12/2018	Zhu	N/A	N/A
10469753	12/2018	Yang	N/A	N/A
10473763	12/2018	Schwarz	N/A	N/A
10473767	12/2018	Xiang	N/A	N/A
10473768	12/2018	Walsh	N/A	N/A
10473770	12/2018	Zhu	N/A	N/A
10473784	12/2018	Puglia	N/A	N/A
10474160	12/2018	Huang	N/A	N/A
10474161	12/2018	Huang	N/A	N/A
10481267	12/2018	Wang	N/A	N/A
10481268	12/2018	Vlaiko	N/A	N/A
10482740	12/2018	Fang	N/A	N/A
10488495	12/2018	Sebastian	N/A	N/A
10488496	12/2018	Campbell	N/A	N/A
10488497	12/2018	Cheong	N/A	N/A
10491052	12/2018	Lenius	N/A	N/A
10491855	12/2018	Gates	N/A	N/A
10495757	12/2018	Dussan	N/A	N/A
10502813	12/2018	Schultz	N/A	N/A
10503174	12/2018	Lim	N/A	N/A
10503175	12/2018	Agarwal	N/A	N/A
10509111	12/2018	Park	N/A	N/A
10509112	12/2018	Pan	N/A	N/A
10509120	12/2018	Bilik	N/A	N/A
10509198	12/2018	Zhou	N/A	N/A
10514444	12/2018	Donovan	N/A	N/A
10514447	12/2018	Schwarz	N/A	N/A
10520591	12/2018	Kotelnikov	N/A	N/A
10520592	12/2018	Droz	N/A	N/A
10520602	12/2018	Villeneuve	N/A	N/A
10523880	12/2018	Gassend	N/A	N/A

10527726	12/2019	Bartlett	N/A	N/A
10531004	12/2019	Wheeler	N/A	N/A
10534074	12/2019	Slobodyyanyuk	N/A	N/A
10534079	12/2019	Kim	N/A	N/A
10539116	12/2019	Davoust	N/A	N/A
10539661	12/2019	Hall	N/A	N/A
10539663	12/2019	Liu	N/A	N/A
10545222	12/2019	Hall	N/A	N/A
10545238	12/2019	Rezk	N/A	N/A
10545240	12/2019	Campbell	N/A	N/A
10545289	12/2019	Chriqui	N/A	N/A
10551501	12/2019	LaChapelle	N/A	N/A
10552691	12/2019	Li	N/A	N/A
10556585	12/2019	Berger	N/A	N/A
10557923	12/2019	Watnik	N/A	N/A
10557924	12/2019	Jang	N/A	N/A
10557926	12/2019	Gilliland	N/A	N/A
10557927	12/2019	Marron	N/A	N/A
10557929	12/2019	Kajiyama	N/A	N/A
10557939	12/2019	Campbell	N/A	N/A
10557940	12/2019	Eichenholz	N/A	N/A
10557942	12/2019	Belsey	N/A	N/A
10564261	12/2019	Huebner	N/A	N/A
10564263	12/2019	Efimov	N/A	N/A
10564266	12/2019	O'Keeffe	N/A	N/A
10564285	12/2019	Belsley	N/A	N/A
10565457	12/2019	Luo	N/A	N/A
10571552	12/2019	Gao	N/A	N/A
10571567	12/2019	Campbell	N/A	N/A
10571570	12/2019	Paulsen	N/A	N/A
10571574	12/2019	Yavid	N/A	N/A
10571683	12/2019	Low	N/A	N/A
10576011	12/2019	Krishnan	N/A	N/A
10578717	12/2019	Bucina	N/A	N/A
10578719	12/2019	O'Keeffe	N/A	N/A
10578720	12/2019	Hughes	N/A	N/A
10578721	12/2019	Jang	N/A	N/A
10578724	12/2019	Droz	N/A	N/A
10578742	12/2019	Guo	N/A	N/A
10585174	12/2019	Gnecchi	N/A	N/A
10585175	12/2019	Reterath	N/A	N/A
10591598	12/2019	Jeong	N/A	N/A
10591599	12/2019	O'keeffe	N/A	N/A
10591600	12/2019	Villeneuve	N/A	N/A
10591601	12/2019	Hicks	N/A	N/A
10591604	12/2019	Xu	N/A	N/A
10591740	12/2019	McMichael	N/A	N/A
10598769	12/2019	Rodrigo	N/A	N/A
10598770	12/2019	Singer	N/A	N/A
10598788	12/2019	Dussan	N/A	N/A

10598791	12/2019	Jain	N/A	N/A
10598922	12/2019	Low	N/A	N/A
10600930	12/2019	Suzuki	N/A	N/A
10605899	12/2019	Singer	N/A	N/A
10605900	12/2019	Spuler	N/A	N/A
10605918	12/2019	Wong	N/A	N/A
10605924	12/2019	Slutsky	N/A	N/A
RE47942	12/2019	Hall	N/A	N/A
D882430	12/2019	Haban	N/A	N/A
10613200	12/2019	Hallstig	N/A	N/A
10613201	12/2019	Pacala	N/A	N/A
10613204	12/2019	Warke	N/A	N/A
10613224	12/2019	Jeong	N/A	N/A
10620301	12/2019	Wilton	N/A	N/A
10620302	12/2019	Zhu	N/A	N/A
10620315	12/2019	Zellinger	N/A	N/A
10620317	12/2019	Chai	N/A	N/A
10620318	12/2019	Yi	N/A	N/A
10627490	12/2019	Hall	N/A	N/A
10627491	12/2019	Hall	N/A	N/A
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Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS (1) This application claims priority on U.S. Provisional Patent Application Ser. No. 63/155,169, filed on Mar. 1, 2021; and this application is also a continuation in part of U.S. application Ser. No. 17/356,676, filed on Apr. 3, 2021, having the title “Hybrid LIDAR with Optically Enhanced Scanned Laser,” which claims priority on U.S. Provisional Application Ser. No. 63/154,990, filed on Mar. 1, 2021; this application is also a continuation in part of U.S. application Ser. No. 17/000,464, filed on Aug. 24, 2020, which claims priority on U.S. Provisional Application Ser. No. 62/890,189, filed on Aug. 22, 2019; and this application is also a continuation in part of U.S. application Ser. No. 16/744,410, filed on Jan. 16, 2020, which is a continuation of U.S. application Ser. No. 15/432,105, filed on Feb. 14, 2017, which claims priority on U.S. Provisional Patent Application Ser. No. 62/295,210, filed on Feb. 15, 2016; and all disclosures of these applications are incorporated herein by reference.

FIELD OF THE INVENTION
(1) The present invention relates generally to the field of optical Time-of-Flight (ToF) measurement, and more specifically, to directionally-resolved ToF measurement technology, known as Laser Radar (LADAR).
BACKGROUND OF THE INVENTION
(2) In the modern era, beginning in World War II, radio detection and ranging (RADAR) systems were deployed, and which utilize reflected radio waves to identify the position of enemy aircraft. Sonar similarly uses sound waves to locate vessels within the oceans. Soon after the development of laser technologies, light/laser detection and ranging (LIDAR/LADAR) systems underwent development.
(3) LADAR is generally based on emitting short pulses of light within a certain Field-Of-View (FOV) at precisely-controlled moments, collecting the reflected light and determining its Time-of-Arrival, possibly, separately from different directions. Subtraction of the pulse emission time from

ToA yields ToF, and that, in turns, allows one to determine the distance to the target the light was reflected from. LADAR is the most promising vision technology for autonomous vehicles of different kinds, as well as surveillance, security, industrial automation, and many other areas, where detailed information about the immediate surroundings is required. While lacking the range of radar, LADAR has a much higher resolution due to much shorter wavelengths that are used for sensing, and hence, comparatively relaxed diffractive limitations. It may be especially useful for moving vehicles, both manned, self-driving, and unmanned, if it could provide detailed 3D information in real-time, with the potential to revolutionize vehicles' sensing abilities and enable a variety of missions and applications.

(4) However, until recently, LADARs have been prohibitively large and expensive for vehicular use. They were also lacking in desirable performance: to become a true real-time vision technology, LADAR must provide high-resolution imagery, at high frame rates, comparable with video cameras, in the range of 15-60 fps, and cover a substantial solid angle. Ideally, a LADAR with omnidirectional coverage of 360° azimuth and 180° elevation would be very beneficial. Collectively, these requirements may be called “real-time 3D vision”.

(5) A variety of approaches has been suggested and tried, including mechanical scanning, non-mechanical scanning, and imaging time-of-flight (ToF) focal-plane arrays (FPA). There is also a variety of laser types, detectors, signal processing techniques, etc. that have been used to date, as shown by the following.

(6) U.S. Patent Application Pub. No. 2012/0170029 by Azzazy teaches 2D focal plane array (FPA) in the form of a micro-channel plate, illuminated in its entirety by a short power pulse of light. This arrangement is generally known as flash LADAR.

(7) U.S. Patent Application Pub. No. 2012/0261516 by Gilliland teaches another embodiment of flash LADAR, with a two dimensional array of avalanche photodiodes illuminated in its entirety as well.

(8) U.S. Patent Application Pub. No. 2007/0035624 by Lubard teaches a similar arrangement with a 1D array of detectors, still illuminated together, while U.S. Pat. No. 6,882,409 to Evans further adds sensitivity to different wavelengths to flash LADAR. Another approach is the use a 2D scanner and only one detector receiving reflected light sequentially from every point in the FOV, as taught by US Patent Application Pub. No. 2012/0236379 by da Silva.

(9) Additional improvements to this approach are offered by U.S. Pat. No. 9,383,753 to Templeton, teaching a synchronous scan of the FOV of a single receiver via an array of synchronized MEMS mirrors. This arrangement is known as retro-reflective.

(10) Yet another approach is to combine multiple lasers and multiple detectors in a single scanned FOV, as taught by U.S. Pat. No. 8,767,190 to Hall.

(11) See also, the laser range camera of U.S. Pat. No. 5,870,180 to Wangler.

(12) The herein disclosed apparatus provides improvements upon prior art LADAR systems.

(13) It is noted that the citing of any reference within this disclosure, i.e., any patents, published patent applications, and non-patent literature, is not an admission regarding a determination as to its availability as prior art with respect to the herein disclosed and claimed method/apparatus.

OBJECTS OF THE INVENTION

(14) The present invention is aimed at overcoming the limitations of both scanning and imaging approaches to LADAR by combining their advantages and alleviating their problems, including but not limited to: 1. Making a light origin point for a LIDAR system at the side of the sensor array, and using a spot, possibly being oval, that may bleed into the imaging field, to eliminate parallax; 2. Prevent the emitted light from the LIDAR system of a vehicle from being bounced off an object that is suddenly nearer than objects that were previously being imaged, to prevent damages to imaging sensors; and 3. Create one or more imaging sensor arrangement that will prevent damage thereto when being exposed to high intensity laser light from the LIDAR system of another vehicle or any other source.

(15) Further objects and advantages of the invention will become apparent from the following description and claims, and from the accompanying drawings.

SUMMARY OF THE INVENTION

(16) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

(17) For a LIDAR system, it is disadvantageous to have the optical reception path and the optical transmission path for the laser beam of the system being completely identical (e.g., being co-axial), as objects (e.g., a mirror) in the transmission path would block light along the reception path. Plus the LIDAR system needs to "see" (i.e., image) from between one meter to 200 meters, and account for parallax. LIDAR system embodiments disclosed herein are particularly configured to address these and other issues.

(18) In accordance with at least one embodiment of the herein disclosed apparatus, an optical system for a light detection and ranging (LIDAR) system may broadly include a laser, an optical transmission system, and an optical reception system. The laser is configured to emit a beam of light at a desired wavelength or wavelengths.

(19) The optical transmission system is configured to shape the beam of light emitted by the laser, and may include: a first mirror and a fold mirror. The first mirror is configured to pivot about an axis to scan the beam of light, and the fold mirror is positioned to reflect the scanned beam of light along a fan of transmission light paths toward a target.

(20) The optical reception system is configured to collect the laser light reflected from the target along a fan of reception light paths, and may include: a lens, and a sensor array that may have a field of view (FOV) on at least a portion of the target.

(21) To improve the performance of the LIDAR system the optical transmission system may also include a wedge-shaped optical element that is positioned to create an auxiliary field of view. The wedge-shaped optical element is positioned proximate to an outside edge of the lens, and the fold mirror is positioned proximate to, but beyond the outside edge of the lens. The wedge-shaped optical element may be formed to have a width that covers one-quarter of the lens. The wedge-shaped optical element may also be formed with a variable transparency, having increasing transparency with increasing distance from a center of the first lens, thereby preventing excessive signals from being received at the linear sensor array when the target is extremely close and therefore the reflected signal is very strong.

(22) The fold mirror may just be a single mirror, and in another embodiment the functionality of the fold mirror may be achieved by a plurality of mirrors arranged in an arcuate pattern to reflect the scanned beam of light toward the target.

(23) Several different embodiments of a scanner may be used. In one embodiment the scanner may be a mirror configured to pivot about an axis to scan the beam of light toward the fold mirror. In another embodiment, the scanner may be a polygonal reflector configured to rotate about an axis to sequentially scan the beam of light toward each of the plurality of mirrors arranged in the arcuate pattern.

(24) Several different embodiments of the sensor array may be utilized. In one embodiment the sensor array may be formed of a plurality of sensors each having a unique length and width, with a center of each of the plurality of sensors being arranged in a line to form a linear sensor array. In another embodiment, the sensor array may be formed of a plurality of sensors each having the same length and width, with a lengthwise direction of each of the plurality of sensors are parallel, but a center of each of the plurality of sensors being positioned to form an arcuate shape, to compensate for an arcuate scan line. In yet another embodiment, the sensor array may be formed to include a first plurality of sensors and a second plurality of sensors. The first plurality of sensors each have a first length and a first width, with a center of each of the first plurality of sensors being arranged in

a line to form a first column of sensors; and the second plurality of sensors each have a longer length and the same width, with a center of each of the second plurality of sensors being arranged in a line to form a second column of sensors. Each of the first plurality of sensors are configured to have a field of view in focus at far field positions; and each of the second plurality of sensors are configured to have a field of view in focus at near field positions, to correct for parallax effect/error. In yet a further embodiment, the sensor array may be formed to include a first plurality of sensors, and a plurality of guard rail sensors. The first plurality of sensors each have the same length and width, with a center of each of the first plurality of sensors being arranged in a line to form a linear array of the sensors, The plurality of guard rail sensors are smaller and are configured to have a lower sensitivity and a higher immunity to damage from stray laser light than each of the first plurality of sensors. The guard rail sensors are positioned to surround the first plurality of sensors, so that corresponding electronic circuitry may be configured to shut down the first plurality of sensors when any one or more of the guard rail sensors receive laser light above a threshold intensity that be harmful.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The description of the various example embodiments is explained in conjunction with appended drawings, as follows.
(2) FIG. **1** illustrates a concept for a 1D scanning and imaging hybrid system.
(3) FIG. **2**A illustrates a 1D hybrid LADAR installed on a Unmanned Aerial Vehicle (UAV) to obtain 2D ToF data.
(4) FIG. **2**B illustrates a 1D hybrid LADAR coupled to an additional scanning mirror to obtain 2D ToF data.
(5) FIG. **2**C illustrates a 1D hybrid LADAR installed on a rotating platform to obtain 2D ToF data.
(6) FIG. **3**A illustrates tight focusing of the laser beam on the scanning mirror of a LADAR and subsequent re-collimation with a cylindrical lens.
(7) FIG. **3**B is a bottom view of the LADAR of FIG. **3**A.
(8) FIG. **3**C is a side view of the LADAR of FIG. **3**A.
(9) FIG. **4** illustrates forming a scan line on the surface of a diffractive element and imaging the scan line onto the target.
(10) FIG. **5**A illustrates a laser spot scanning across virtual detector pixels on the target.
(11) FIG. **5**B illustrates permissible misalignment while the laser spot is scanning across tall virtual detector pixels on the target.
(12) FIG. **5**C illustrates splitting each of the pixels shown in FIG. **5**B into two sub-pixels.
(13) FIG. **5**D illustrates positioning an array of micro-lenses in front of the detectors, and further illustrates how a relatively small pixel can be collecting light from a relatively large virtual pixel.
(14) FIG. **6**A illustrates the laser of the LADAR system being continuously ON.
(15) FIG. **6**B illustrates the laser being modulated with one short pulse per pixel.
(16) FIG. **6**C illustrates the laser being modulated with multiple short pulses per pixel.
(17) FIG. **7** illustrates synchronizing the clock of an electronic control system with the scanning process, so that there is a constant integer number of clock pulses per scanning cycle.
(18) FIG. **8** illustrates the electronic control system of FIG. **7**.
(19) FIG. **9**A illustrates an R-θ-Φ coordinate system, and an X-Y-Z Cartesian coordinate system.
(20) FIG. **9**B illustrates a general layout of a LIDAR system.
(21) FIG. **9**C illustrates a scanned raster pattern in Φ-θ coordinate space.
(22) FIG. **9**D illustrates a scanned laser spot.
(23) FIGS. **9**E and **9**F illustrate aspects of the scanner FOV and the laser FOV.

(24) FIGS. **9**G, **9**H, **9**J and **9**K illustrate changes to the size and position of the laser spot versus the target distance.
(25) FIG. **10** is a perspective view of one embodiment of an optical system usable in a LIDAR system as disclosed herein, in which the light origin point is directly centered on the target and centered on the sensor array, which may result in parallax.
(26) FIG. **10**A is the perspective view of FIG. **10**, but is shown with the center of the laser-scanned line being offset from the field of view of the sensor with respect to the target.
(27) FIG. **10**B is a top view of the optical system as shown in FIG. **10**A.
(28) FIG. **10**C shows a portion of the optical system of FIG. **10**B, in which the fold mirror is moved to a position in front of the lens and is appropriately angled so that the scanned laser beam remains parallel to sensor field of view.
(29) FIG. **10**D shows the portion of the optical system seen in FIG. **10**C, but where the fold mirror is angled so that the scanned laser beam converges with the sensor field of view.
(30) FIG. **10**E shows the portion of the optical system seen in FIG. **10**D, but where the fold mirror is positioned well outside the field of view of the lens, and the linear sensor array is also positioned offset relative to the normal vector of the lens.
(31) FIG. **10**F shows a portion of the optical system of FIG. **10**B, in which an additional optical element, such as a negative cylindrical lens, is positioned towards the outside edge of the main lens.
(32) FIG. **10**G shows a top view of the main lens and the additional optical element of FIG. **10**F.
(33) FIG. **10**H illustrates the portion of the optical system seen in FIG. **10**B, in which the linear sensor array is positioned for light reflected from the main target is focused by the main lens upon the linear sensor array, and light reflected off of a secondary target passes through the main lens, and is defocused at the linear sensor array but is focused at a distal location.
(34) FIG. **10**J an embodiment where the sensor is both extended and tilted in the direction of the focused image from the nearby target, such that the light from the target at any range reaches the sensor in a nearly-focused state.
(35) FIG. **10**K is an alternative embodiment to that shown in FIG. **10**G, wherein the additional optical element has a variable transparency shown by series of lines therein.
(36) FIGS. **11**Ai-**11**Aiii are a series of orthogonal views showing an unscanned laser beam, a scanning mirror **104**, the scanned laser beam, and the resulting laser scan line on the target, where the unscanned laser beam **103***a* falls solely within the Z-Y plane, where the laser scan line on the target is linear and matches a linear sensor FOV.
(37) FIGS. **11**Bi-**11**Biii are a series of orthogonal views showing an unscanned laser beam that has a Y-plane offset, a scanning mirror, a scanned laser beam, and the resulting laser scan line on target.
(38) FIG. **11**C is an embodiment that is the same as shown in FIG. **11**B, but which includes an additional optical element with variable curvature to straighten the curved smiley shaped scan line.
(39) FIG. **11**D illustrates the scanning arrangement of FIG. **11**C, but where an additional optical element with variable curvature is placed between the lens and the target to correct for the curvature of the scan line transmitted by a lens, to form a curved scan line.
(40) FIG. **11**E illustrates the scanning arrangement of FIG. **11**B, but where the pixels on the sensor are arranged in a curved pattern to match the curvature of the received reflection of a curved shaped scan line.
(41) FIG. **12**Ai shows such a cross-section through a mirror having a handle layer and a device layer, with a photo resist mask having been deposited at locations where the handle layer is to remain intact, so that an applied etchant can remove select portions of the handle layer.
(42) FIG. **12**Aii shows the resulting structure after the etchant has removed portions of the mirror of FIG. **12**Ai.
(43) FIG. **12**Bi shows a cross-sectional view of a mirror that has been formed out of a single layer silicon wafer, with an oxide mask having been applied to portions thereof where the silicon wafer is

to remain intact.
(44) FIG. **12**Bii shows the mirror cross-section of FIG. **12**Bi after an applied etchant has removed portions of the silicon wafer.
(45) FIG. **12**Ci shows the mirror cross-section of FIG. **12**Bii being masked prior to receiving an etchant to undergo processing a second time.
(46) FIG. **12**Cii shows the mirror cross-section of FIG. **12**Ci after the etchant has removed additional portions of the silicon wafer.
(47) FIG. **12**D is an image of a scan mirror after an etchant has removed portions thereof.
(48) FIG. **12**E is a view of an improved mirror arrangement that includes torsional hinges that connect to a flexure that is mounted to piezoelectric elements.
(49) FIG. **12**F illustrates another improved mirror embodiment, in which torsional hinges taper down toward the mirror active surface.
(50) FIG. **12**G shows a cross section through the hinge of one of the mirrors of FIGS. **12**E and **12**F, illustrating the formation of microcracks in the high-stress silicon MEMS structure.
(51) FIG. **12**H illustrates a hinge embodiment adapted to reduce the microcracks, which includes application of an oxidization layer over the structure, including the microcrack.
(52) FIG. **12**J shows the hinge of FIG. **12**H after the oxidization layer is removed, showing a reduction in size of the treated microcrack.
(53) FIG. **12**K is a cross section through the mirror of FIGS. **12**E and **12**F, but where a high-reflectivity dielectric coating has been applied, which bows the mirror.
(54) FIG. **12**L is a cross section through the mirror of FIGS. **12**K, but where a stress-relief sub-layer such as chromium has been deposited between the high-reflectivity dielectric coating and the mirror, which bows the mirror in the opposite direction, and cancels out the bowing from the high-reflectivity dielectric coating.
(55) FIGS. **12**M, **12**N, **12**P, and **12**Q illustrate different ways to drive the scan mirror;
(56) FIG. **12**R illustrates a laser projecting an unscanned beam towards a mirror which scans the beam towards mirrors that are arranged to reflect the beam towards a target and form a scan line thereon.
(57) FIG. **12**S illustrates a laser projecting an unscanned beam towards a rotating polygon which scans the beam towards mirrors that are arranged to reflect the beam towards a target and form a scan line thereon.
(58) FIG. **12**T illustrates an arrangement in which the mirror scans a laser beam through a telescopic lens elements having power either one or two dimensionally.
(59) FIGS. **13**A-**13**H show various embodiments for generating a slow speed scan stage for an optical system of the LIDAR disclosed herein.
(60) FIGS. **14**A through **14**Dii show various embodiments of sensor arrays for an optical system of the LIDAR disclosed herein.
(61) FIG. **14**E is a diagram illustrating that each individual sensor shown in FIG. **14**Di and FIG. **14**Dii may be connected to a trans-impedance amplifier that is connected to a threshold comparator which is connected to a time stamp circuit.
(62) FIG. **14**F is a diagram illustrating an embodiment similar to the one shown in FIG. **14**E, except that a second threshold comparator and a second time stamp are added to the processing channel for each APD pixel.
(63) FIG. **14**G is a diagram illustrating an embodiment similar to the one shown in FIG. **14**E, except that the amplified signals from pixels are not just compared to one or more preset threshold values, but are also digitized as sequences of samples using analog-to-digital converters.
(64) FIG. **15**A illustrates circular and non-astigmatic light from a fiber laser passing through an aspheric lens and then being scanned by mirror.
(65) FIG. **15**Bi and FIG. **15**Bii are top and side views illustrating light from a MOPA laser passing through a cylindrical lens for fast direction MOPA collimation and then passing through a

cylindrical lens for slow direction MOPA collimation, and then being scanned by mirror.
(66) FIG. **15**Ci and FIG. **15**Cii illustrate another embodiment for focusing a MOPA laser with one aspheric and one cylindrical lens.
(67) FIG. **15**Di and FIG. **15**Dii illustrate an embodiment similar to the one shown in FIG. **15**Bi and FIG. **15**Bii, but which additionally includes a low-reflectance signal sampling mirror that diverts some of the light beam coming out from the cylindrical lens.
(68) FIGS. **15**Ei and **15**Eii illustrate the addition of a dynamic sensor that may be used to detect astigmatism, and can adjust the signal.
(69) FIG. **16**A illustrates an optical arrangement that may be used for accurate control of the motion of a high-speed scanning mirror, and which uses a pickoff laser and two pickoff sensors.
(70) FIG. **16**B shows the mirror scan profile and the pulses from the end pickoff sensors and the pulses for the center pickoff sensors of the embodiment of FIG. **16**A.
(71) FIG. **16**C is the same a FIG. **16**B, but also includes a digital representation of the laser on interval and the laser off interval.
(72) FIG. **16**D is a piezo drive that includes a control circuit connected to a resonant power driver, which connects through an inductor to the piezoelectric element, which in turn drives the mirror.
(73) FIG. **16**E illustrates a biased drive circuit that may be used to drive the piezoelectric elements that cause oscillation of the mirror of one of the herein disclosed optical scanning systems.
(74) FIG. **16**F is a drive circuit configured such that different piezo elements used to drive the mirror to oscillate are driven at different phases.
(75) FIG. **16**G is diagram showing a drive circuit that is configured such that sensor timing is driven based upon the end pickoff pulse.
(76) FIG. **16**H is diagram showing an arrangement where a PLL is utilized to smooth the pickoff pulse in the event that any jitter occurred in the pickoff pulse.
(77) FIGS. **17**A-**17**E illustrate optical arrangements where the horizontal slow stage of scanning is implemented by rotating a high-speed linear scanned column.
(78) FIG. **17**F shows a car that uses mounting structure to support a LIDAR system as disclosed herein.
(79) FIG. **18** illustrates a spread sheet that represents the position and timing of two optical pick-off sensors.
DETAILED DESCRIPTION OF THE INVENTION
(80) As used throughout this specification, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than a mandatory sense (i.e., meaning must), as more than one embodiment of the invention may be disclosed herein. Similarly, the words “include”, “including”, and “includes” mean including but not limited to.
(81) The phrases “at least one”, “one or more”, and “and/or” may be open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “one or more of A, B, and C”, and “A, B, and/or C” herein means all of the following possible combinations: A alone; or B alone; or C alone; or A and B together; or A and C together; or B and C together; or A, B and C together.
(82) Also, the disclosures of all patents, published patent applications, and non-patent literature cited within this document are incorporated herein in their entirety by reference. However, It is noted that the citing of any reference within this disclosure, i.e., any patents, published patent applications, and non-patent literature, is not an admission regarding a determination as to its availability as prior art with respect to the herein disclosed and claimed apparatus/method.
(83) Furthermore, any reference made throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection therewith is included in at least that one particular embodiment. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Therefore, the described

features, advantages, and characteristics of any particular aspect of an embodiment disclosed herein may be combined in any suitable manner with any of the other embodiments disclosed herein.
(84) Additionally, any approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative or qualitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term such as “about” is not to be limited to the precise value specified, and may include values that differ from the specified value in accordance with applicable case law. Also, in at least some instances, a numerical difference provided by the approximating language may correspond to the precision of an instrument that may be used for measuring the value. A numerical difference provided by the approximating language may also correspond to a manufacturing tolerance associated with production of the aspect/feature being quantified. Furthermore, a numerical difference provided by the approximating language may also correspond to an overall tolerance for the aspect/feature that may be derived from variations resulting from a stack up (i.e., the sum) of a multiplicity of such individual tolerances.
(85) In accordance with at least one embodiment of the hybrid LADAR system disclosed herein, a fast 1D scanner/imager hybrid can collect one line of high-resolution ToF data within tens of microseconds. If needed, that hybrid can also be coupled to a secondary, slow scanning stage, producing raster ToF frames of thousands of lines at high frame rates, with total pixel throughput of tens or even hundreds of megapixels per second.
(86) The following description lists several embodiments of the present invention, which are merely exemplary of many variations and permutations of the subject matter disclosed.
(87) Mention of one or more representative features of a given embodiment is likewise exemplary: an embodiment can exist with or without a given feature, and likewise, a given feature can be part of other embodiments.
(88) A preferable embodiment of a 1D hybrid scanner/imager is illustrated in FIG. **1**. It comprises a 1D array of photo-detectors **1**, placed in a focal point of a receiving optical system **2**, and a scanning mirror **3**, its axis of rotation perpendicular to the direction of the detector array. A laser beam **5** generated by the laser **4** is directed onto the scanning mirror **3** through a collimating lens **6**. The positions of the optical system **2** and the scanner **3** are adjusted in such way that the fan of imaginary rays **7** emanating from individual pixels of the array, and the fan of real laser rays **8** emanating from the scanning mirror **3** lay in the same plane. Respectively, the laser scan line on the target overlaps with the FOV of the detector array.
(89) Preferably, the scanning mirror **3** is a resonant MEMS mirror. Such mirrors, having dimensions of the reflective area of the order of 1 sq. mm, the resonant frequencies of tens of kHz, and the scan amplitude of tens of degrees, are becoming commercially available. While generally this invention would benefit from the fastest rate of scanning, a general tendency in scanning mirrors is that the faster scanning rate typically leads to narrower scan angle and smaller mirror size, which in turn increases the divergence of the scanned laser beam, thus limiting the number and size of the detectors in the array, and the amount of light that can be collected onto the detectors, especially, at longer ranges.
(90) This invention might further benefit from non-mechanical beam scanners (NMBS) that are undergoing development, although their specifications and commercial availability remain unclear. For example, U.S. Pat. No. 8,829,417 to Krill teaches a phase array scanner, and U.S. Pat. No. 9,366,938 to Anderson teaches an electro-optic beam deflector device. NMBS may allow scan rates, or beam cross-sections, exceeding those of mechanical scanners.
(91) A photo-detector array should preferably consist of high-sensitivity detectors, as the number of photons arriving to each detector, especially from longer ranges, might be exceedingly small. Significant progress has been recently achieved in design and manufacturing of Avalanche Photo Diodes (APD), working in both a linear, and a Geiger mode, which is also known as a Single-Photon Avalanche Detector (SPAD). The former are reported to be able to detect a light pulse

consisting of just a few photons, while the latter can actually be triggered by a single photon, and both types would be suitable for embodiments of the present invention. It should be noted, however, that the most advanced detectors are expensive to fabricate, therefore, the present invention that needs only one row of detectors would be very cost-efficient in comparison with flash LADAR that would typically require a 2D array of the same resolution.
(92) One of the problems typically encountered in ground-based and vehicular LADARs intended to be used in a populated area is the eye safety hazard presented by its powerful lasers. Therefore, a preferable embodiment of this invention would use a laser with longer infra-red (IR) wavelength, exceeding approximately 1400 nm. Such longer wavelengths are not well-absorbed by human eyes and therefore don't pose much danger. Respectively, much more powerful lasers might be used in this spectral range.
(93) Depending on a particular task the hybrid LADAR is optimized for, one of the following configurations of the slow stage may be used for ground surveillance, vehicular navigation and collision avoidance, control of industrial robots, other applications, or alternatively there may not be any slow stage at all. As depicted on FIG. **2**A, the hybrid 1D LADAR of FIG. **1** is attached to an aerial platform **15**, with the scan direction perpendicular to direction of platform motion.
(94) FIG. **2**B illustrates using a scanning mirror **16**, with its scanning direction being perpendicular to the scanning direction of the hybrid. Combined, they constitute a 2D LADAR preferably having comparable scan angles in both directions. Finally, FIG. **2**C illustrates positioning the hybrid of FIG. **1** on a rotational stage **17**, which gives the combined LADAR a 360 degree fOV in a horizontal plane.
(95) If a scanning mirror is used for the slow stage, its area should be sufficiently large to reflect the entire fan of beams emanating from the fast scanner, as well as the entire fan of rays coming to the optical reception system. However, scanning mirrors with active area of square centimeters, frequencies of tens of Hz, and scan angles of tens of degrees are commercially available. Thus, an exemplary combination of a detector array with 300 pixels, a fast scanner of 30 kHz, and a slow scanner of 30 Hz would provide a point cloud of ˜**18**M points a second, assuming that both scanners utilize both scan directions, thus enabling real-time 3D vision with the resolution of approximately 1000×300 pixels at 60 frames a second. Rotation at 60 revolutions per second, or linear motion on board a vehicle would also provide comparable point acquisition rates, without the limitations of the reflective area of the scanning mirror.
(96) Preferably, the optical system **2** is configured to form the FOV of each individual pixel **11** of the detector array to match the divergence angle of the laser beam, which is generally possible and usually not difficult to achieve. In an exemplary embodiment, a laser beam of near-infrared (NIR) light of approximately 1 mm in diameter will have a far-field divergence of approximately 1 milliradian (mrad). A photo-detector pixel 20 um in size will have similar FOV when placed in a focal plane of a lens with 20 mm focal length.
(97) It might also be advantageous to use additional elements in an optical transmission system to shape the laser beam, such as, for example, a cylindrical lens **23**, as depicted in FIG. **3**A. In this case, the laser may still be nearly-collimated in the scanning direction, but may be tightly focused in the orthogonal direction, as illustrated by FIG. **3**B, which shows the view from direction B in FIG. **3**A. Cylindrical lens **23** subsequently re-collimates the beam in the orthogonal direction as illustrated on FIG. **3**C, which shows the view from direction C in FIG. **3**A. The advantage of such an arrangement lies in the possibility to reduce the dimension of the scanning mirror in the tightly-focused laser beam direction, as well as in reduced eye hazard presented by the laser beam expanded in at least one direction.
(98) In another embodiment of the present invention, the laser beam can be tightly focused in both directions and be scanned across a diffractive element **21**, as illustrated on FIG. **4**, forming a thin scan line on its surface. That scan line is subsequently imaged onto the target by the lens **20**. The lens **20** may be identical to the lens **2** in the optical reception system in front of the detector array,

in which case the dimensions of the scan line on the surface of a diffractive element **21** should be similar to that of the detector array, thus insuring the equal divergence of both FOVs of the illuminating laser and the detector. Alternatively, the lenses **2** and **20** may have different magnification, and respectively, the scan line may have different dimensions from the detector array, however, the design should provide for a good overlap of both FOVs on the target, as further illustrated on FIG. **5**A, where virtual pixels **31** denote the projections of the detector pixels **11** onto the target through the optical system **2**, while the laser spot **25** scans the same target along the scan line **22**, thus sequentially illuminating pixels **11** with light reflected from the target. As shown on the figure for clarity, at the moment when the laser spot assumes position **25***a*, it illuminates the detector pixel **11***a*. It should be noted that the desired shape of laser spot **25** may be achieved by a variety of optical methods, as illustrated on FIG. **3** and FIG. **4**, or other methods, without limiting the scope of this invention.
(99) Pixels **11** may have different shapes: for example, FIG. **5**B illustrates pixels with their height exceeding the pitch of the array. For example, pixels of the array may be 20 micrometers wide and 60 micrometers toll. The advantage of such arrangement is that it may accommodate slight misalignment between the laser scan line **22** and the line of virtual pixels **31**. Likewise, the laser spot may also be oblong, or have some other desirable shape. It might also be advantageous to split each of the pixels **11** into two or more sub-pixels **11***b*, **11***c*, as depicted on FIG. **5**C. These sub-pixels would generate redundant information, as the scan spot **25** may cross either of virtual pixels **31***b*, or **31***c*, or both of them, at the same time, and the ToA measured by either sub-pixels or both would be treated as one data point. The advantage, however, may come from the fact that smaller sub-pixels would generally have lower noise level, therefore a laser spot of the same power is more likely to generate a response when illuminating a small sub-pixel, than a larger full pixel.
(100) Some types of the high-sensitivity detectors, notably Geiger-mode APDs, are known to need considerable gaps between active pixels to eliminate cross-talk, thus limiting the fill factor of the arrays consisting of such detectors. To alleviate this problem, an array of micro-lenses **12** may be used in front of the detectors **11**, as depicted on FIG. **5**D, with the micro-lens **12***a* specifically illustrating how a relatively small pixel **11***a* can still be collecting light from a relatively large virtual pixel **31***a*.
(101) It is also preferable to match the total extent of the array's FOV with the total scan angle, for example: 512 pixels, 20 um each, placed behind a 20-mm lens will subtend the angle of ˜29°. Respectively, the total scan angle should be the same or slightly greater to utilize every pixel of the array.
(102) In a further embodiment of this invention, as the laser spot moves along the scan line, it is continuously energized, thus producing a time-domain response in the pixels it crosses, as illustrated on FIG. **6**A, where the graph **10**A represents the intensity (I) vs. time (t) response in the pixel **11***a*. The response starts when the laser beam in position **25***b* just touches the virtual pixel **31***a*, and ends when the laser beam moves to position **25***c*, just outside of the virtual pixel **31***a*. In such an embodiment, the laser power, and hence, the sensitivity is maximized, however, the precision of ToF measurement may suffer, due to the ambiguity of the Time of Arrival (ToA) of a relatively long light pulse.
(103) To alleviate this problem, the laser can be electrically-modulated with shorter pulses, as illustrated by FIG. **6**B. For example, the laser is turned on when the laser spot is in the position **26***d*, producing the response **10***d* in the pixel **11***a*. Since the pulse is narrower, its ToA can be determined with greater precision.
(104) One problem with this modulation method might arise if the laser is energized while its spot falls in between two virtual pixels, therefore each of the real pixels is getting only a fraction of the reflected light. Due to some parallax between the laser scanner and the detector array, the precise overlap between the laser spot and virtual pixels is somewhat dependent on the distance to the target, and hence not entirely predictable. To alleviate this problem, the laser may be modulated

with more than one pulse per pixel, as illustrated on FIG. **6**B. In this case, at least one of the consecutive modulation pulses emitted with the laser spot in positions **25***f*, **25***g*, **25***h*, would fully overlap with the virtual pixel **31***a*, thus producing full response **10***g* in the pixel **11***a*. Responses from two other pulses **10***f* and **10***h* might be shared with adjacent pixels and hence be lower.
(105) In any LIDAR system dependent on ToF measurements, it is important to accurately measure the ToA of a light pulse on the detector, as well as the time when the light pulse was emitted. In the present invention, additionally, it is important to precisely know the position of the scanned spot, as it determines which detector pixel will be illuminated. To achieve this, it is preferable to synchronize the clock of the electronic control system, denoted by pulses **50** on FIG. **7**, with the scanning process, in such a way that there is a constant integer number of clock pulses per scanning cycle **51**. The vertical axis(S) on FIG. **7** illustrates a scan angle, while the horizontal axis (t) illustrates time. Since many resonant scanners have unique, non-tunable resonant frequency, the system clock frequency may be changed instead to keep an integer number of clocks per period, preferably, by means of a Phase-Lock Loop (PLL) circuit. As long as this fixed relationship between the system clock and the scan angle is maintained, the system clock may be easily used for precise detector read-out timing, as well as for timing of the laser modulation pulses, if such modulation is used. Additionally, some types of detectors require quenching after they received a light pulse, and arming, to be able to receive the next pulse. Keeping pixels armed when they are not expected to be illuminated is undesirable, as it carries the risk of false positive due to internal noise. In most flash LADARs with such detectors, they are armed all at once, right before the illuminating laser pulse is emitted. In the present invention, it is preferable to arm detectors one-by-one, in the order they are illuminated by the scanning laser beam. As illustrated on FIG. **7**, while the scanning beam proceeds from bottom to top, the pixel **11***d* would be armed by the system clock pulse **50***d*, while pixel **11***e* will be armed considerably later by the system clock pulse **50***e*.
(106) A preferred embodiment of the electronic control system is illustrated by FIG. **8**. The fast scanning mirror **3** is driven by the driver **63** at the frequency defined by the voltage-controlled oscillator (VCO) **67** through clock divider **65**. Being resonant, mirror **3** tends to oscillate at maximum amplitude while driven at its own resonant frequency and in a specific phase with respect to the drive signal. The mirror phase feedback **71** carries the information about the mirror's phase to the phase-locked loop (PLL) **62** that compares it to the phase of the clock divider and adjusts the VCO to eliminate any phase error, thus maintaining mirror oscillations close to its resonant frequency.
(107) System clock **68** is derived from the same VCO, thus insuring that all other timing blocks function in strict synchronization with the mirror motion. Specifically, the arming circuit **61** provides sequential pixel arming and the laser driver **64** generate laser pulses in specific relationship to the scanned laser spot, as discussed above. Likewise, the same system clock synchronizes the motion of the slow mirror **16** through the driver **66** in a fixed relationship with the motion of the fast mirror, for instance, one cycle of the slow mirror per 1024 cycles of the fast mirror. Read-out circuit **60** and signal processor **69** may use the same clock as well, although they, unlike other elements discussed above, don't have to be strictly synchronized with the mirror motion. They must, however, be fast enough to be able to read and process data from all pixels within one scan cycle of the fast mirror. 3D vision data **70** is generated based on the ToF measurement coming from the detector array **1** and then supplied to users, such as navigation system of autonomous vehicles or security surveillance system.
(108) It should be noted that the present invention offers a considerable advantage in efficiency over other types of LADARs, especially flash LADARs, where short laser pulses are used to illuminate the entire scene at relatively long intervals. The advantage comes from the fact that in the present invention the laser is energized either continuously, or with a fairly high duty cycle: for example, a LADAR of the present invention using 10 ns pulses per pixel and generating **18**M data points per second would have a duty cycle of approximately 35%. A flash LADAR using the same

10 ns pulses, and having the same frame rate of 60 fps, would have the laser duty cycle of only 0.00006%. Consequently, to generate the same average power and attain comparable range, the laser of that hypothetical flash LADAR would need instantaneous power almost 6 orders of magnitude greater. While pulsed lasers capable of producing very short powerful pulses do exist, they are known to have lower efficiency, larger size and higher cost than continuous lasers of the same average power. Aside from inability to deliver high average power, pulsed sources are typically less efficient, more complex, bulkier and costlier, than continuous or high-duty sources. The general physical explanation of lower efficiency is in the fact that the emitted power of photonic sources—lasers or LEDs—is typically proportional to the current, while parasitic losses on various ohmic resistances inside those sources are proportional to the square of the current.

(109) Additionally, high-power pulsed laser sources are typically more dangerous in terms of eye safety.

(110) It should be noted, that it is difficult (although not impossible) to place the scanner at the center of the optical system. At any other position, there will be some parallax between the FOVs of the optical transmission and the reception systems, hence the detailed mapping of the pixels onto a scan line will depend on the distance to the target. However, in a practical system such parallax can be kept to a minimum by placing the scanner in close proximity to the optical system.

(111) The present invention as illustrated by the above-discussed embodiments, would provide serious advantages over other types of LADAR.

(112) In a typical imaging LADAR, the light source emits a short pulse which illuminates all pixels at once. Respectively, each pixel receives only a small fraction of the total back-scattered signal. In the proposed hybrid, only one pixel receives all the back-scattered light emitted at a given moment. If we assumed that the illumination power is the same, then the signal strength on each pixel would be up by a factor comparable to the number of pixels, i.e. hundreds, if not thousands. In practice, pulsed sources generally have higher instantaneous power than continuous ones, but their average power is still considerably lower.

(113) Conversely, in a typical scanning LADAR, at any given moment only a small portion of the photo-detector is receiving any signal, while the rest only generates noise and contributes to unwanted capacitance. The exception is so-called retro-reflective scanners, where a small detector FOV is directed through the same scanning system. However, this approach only works with large, slow scanners, where the mirrors are large enough to provide sufficient optical collection area for back-scattered light. Contrarily, high-speed scanners are usually tiny, just sufficient to fit the beam of the laser, and are usually of the order of 1 mm.

(114) In either case, as illustrated above, a hybrid appears would have a considerable SNR advantage: higher signal than imaging-only, or lower noise than scanning-only device.

(115) It is anticipated that a hybrid LADAR of the present invention will be able to use a regular laser diode as its illumination source—which is by far the cheapest and most efficient source among those suitable for LADARs.

(116) Also, both cost and power consumption are roughly proportional to the total number of pixels fabricated by a given technology. So, substituting a 2D array by a 1D array is supposed to considerably reduce both cost and power consumption for the detector array, while the cost and power consumption of both fast and slow scan stages may be considerably lower, than that of the array of pixels or the illuminating laser.

(117) The above embodiments, which were originally disclosed in Applicant's patent application Ser. No. 15/432,105, now issued as U.S. Pat. No. 10,571,574, describe LIDAR/LADAR systems wherein a laser beam is scanned by a mirror oscillating about a single axis and directed towards a target, and the light reflected off the target is imaged by a 1D sensor array. The scanner quickly scans the laser beam and therefore the reflected image from the target falls on any given pixel in the array for a very short time. This arrangement eliminates the need for fast pulsing of the lasers thereby drastically simplifying the laser.

(118) There are other issues with LIDAR systems that are addressed by the embodiments disclosed hereinafter. However, to better appreciate some of those issues and solutions, which are subsequently described in a “LIDAR Embodiments” section, a “LIDAR Theory of Operation” section is hereinafter provided.
(119) LIDAR Theory of Operation
(120) The polar coordinate system φ-θ, as depicted on FIG. **9**A, will be used in the following description. With respect to the description of LIDAR elements in a given rotational position, Cartesian x-y-z coordinates will be used, and will assume that the z-axis is horizontal and pointing generally in the azimuthal direction of the laser light.
(121) The general layout of the LIDAR system is presented in FIG. **9**B.
(122) The LIDAR system reflects a collimated laser beam off a fast scanning MEMS mirror to form a narrow tall vertical laser field of view (L-FOV) in the x-z plane. A second, much slower pivoting motion sweeps the L-FOV around the x axis.
(123) The laser is continuously on while the beam is scanned within the laser field of view (L-FOV). The laser spot profile is Gaussian with a spot size largely determined by the size of the fast scan mirror.
(124) A linear sensor array and lens combination also forms a narrow tall vertical sensor field of view (S-FOV), which is nearly coincident with the L-FOV. The planes of S-FOV and L-FOV are both vertical, but slightly tilted toward each other, intersecting at a small angle of convergence δ. The heights (i.e. the X-coordinates) of the origin points of both the L-FOV and S-FOV are the same, so there is no vertical parallax between them. However, there is some parallax h between the origin points in the horizontal direction.
(125) Each pixel of the sensor array images a small FOV within the entire S-FOV. If an object is illuminated with the scan beam, it reflects some laser light back towards the sensor. A focusing lens focuses this energy onto one (or more) sensor pixels. In the vertical direction, the L-FOV is slightly larger than S-FOV, thus ensuring that every pixel is illuminated during each laser scan.
(126) Fast Scan (Vertical)
(127) The scan mirror may be a resonant MEMS structure which oscillates at the frequency f.sub.1, and the mechanical amplitude A/2. The optical scan position is twice the mechanical scan position, so the deflected laser beam has the scan amplitude of A. Assuming that in the neutral position of the scan mirror, the laser beam is directed out at the elevation angle θ.sub.0, as a function of time, the elevation angle is:
(128) $(t) = {}_{0} + A * \sin({}_{1} * t - \frac{}{2})$, where ${}_{1} = 2 * \; * f_1$ (1)
The phase offset of π/2 is introduced for convenience, to begin the cycle at t=0, while the scan mirror is in its lowermost position.
The vertical scan velocity is:
(129) ${}'(t) = \frac{d\;(t)}{dt} = A * \; {}_{1} * \cos({}_{1} * t - \frac{}{2})$ (2)
As evident from (2) the scan velocity diminishes toward the ends of the scan line. To avoid scanning across some pixels at very low velocity, the mechanical scan angle is chosen so that the L-FOV is smaller than the total optical scan angle 2*A, and the laser is switched off outside of L-FOV.
(130) The elevation angle of the FOV of each individual pixel of the sensor array is defined by the pixel's position x, and lens's focal distance FL. Assuming that the center of the array, i.e. x=0, corresponds to the same elevation angle θ**0** of the center beam of the scan:
(131) $(x) = {}_{0} - \mathrm{atan}(\frac{x}{FL})$ (3)
(132) Combining (3) and (1), we can determine the time moment when the laser beam is at the same elevation angle as a given pixel, i.e. θ(t)=θ(x):
(133) ${}_{0} + A * \sin({}_{1} * t - \frac{}{2}) = {}_{0} - \mathrm{atan}(\frac{x}{FL})$, or (4) $t(x) = \frac{1}{{}_{1}}(\mathrm{asin}(-\frac{1}{A}\mathrm{atan}(\frac{x}{FL})) + \frac{}{2})$ (5)

(134) Referring to t(x) as Time of Departure (ToD) for a pixel at position x, as the photons emitted from the laser at that moment may eventually reach the pixel x. Photons emitted at any other moment would either reach other pixels, or be lost. Therefore, if the pixel at position x registers a return pulse at t.sub.ToA, the distance to the target d at the elevation θ(x) would be:

$d(\theta)=(t\text{.sub.ToA}-t(x))*c,$ (6)

where c is the velocity of light in air.

(135) It is assumed that a sync pulse will be sent from the mirror control sub-system to the sensor at some known moment, for example, at t=0, i.e., a lowermost point of the scan. This pulse will start the individual pixel ToA counters, that will be stopped by the arrival of the return light pulses. Then, the target distances for each pixel will be calculated using (6).

(136) The scanned laser beam passes through each elevation angle θ twice per cycle: on the way up and on the way down, with its own departure time t(x) in each case. Respectively, each pixel in the array also receives two pulses per cycle, both of which should be processed according to (6). The equation (5), however, for down scan takes the form:

(137) $t(x) = \frac{1}{_1}(\frac{3*}{2} - \text{asin}(-\frac{1}{A}\text{atan}(\frac{x}{FL})))$ (7)

(138) As seen in FIG. **18**, the spreadsheet <Sensor Processing Time_1.xlsx> offers a numerical illustration of the above equations, for S-FOV equal to +/−15 deg. The maximum scan angle of the mirror is +/−20 degrees with the laser being on for slightly more than +/−15 degrees (L-FOV), and off for the rest of the scan. Column D contains Departure Times calculated according to (5) and (7). Column E contains Time of Arrival (ToA) from maximum distance. In reality, each pixel sees a return pulse somewhere in-between ToD and ToA from maximum distance.

(139) As noted, all times are referenced to the lowermost point of the scan.

(140) Note that for every pixel, there is a minimum “quiet” interval between the latest possible ToA during up-scan, and the ToD (and hence, the earliest possible ToA) during subsequent down-scan. This interval is the shortest for the last pixels of the array, being ˜1.3 us.

(141) Additionally, this spread sheet represents the position and timing of two optical pick-off sensors, generating triggers for the fast mirror control. Those sensors are responsible for precise motion control of the fast mirror, and facilitate the generation of the sync pulses, but don't directly affect the LIDAR operation as described above.

(142) Slow Scan (Horizontal)

(143) While the fast mirror scans in vertical direction as described above, a slow rotary stage with frequency f.sub.2 provides continuous constant-velocity motion around the vertical axis X, so the azimuthal angle φ of the vertical plane containing L-FOV can be expressed as:

$\phi(t)=\omega\text{.sub.2}*t$, where $\omega\text{.sub.2}=2*\pi*f\text{.sub.2}$ (8)

The slow rotation is synchronized with the fast scan, so

(144) $f_2 = \frac{f_1}{N},$ (9)

where N is a large integer.

(145) Such synchronization assures a repeatable raster pattern in φ-θ coordinate space, as illustrated on FIG. **9**C. Assuming there is a target within the working range of the LIDAR, each pixel of the sensor receives a return pulse during each scan line, i.e. 2 pulses per each fast mirror cycle.

(146) The scan pattern on FIG. **9**C represents the position of the center of the laser spot. However, the laser beam, while notionally-collimated, has finite divergence due to diffraction, and therefore, forms a spot of some shape and size on the target, which moves as the laser beam scans the target. Then, the lens images such moving spot onto the sensor, where it also moves, as illustrated on FIG. **9**D, forming pulses of finite duration on the pixels it crosses. While the precise shape of those pulses cannot be described analytically, some “semi-qualitative” explanations are below.

(147) First and foremost, the laser spot moves up and down the sensor array with the fast scan, as shown of FIG. **9**D. Since both L-FOV and S-FOV are rotating together and the angle between them does not change, the spot trajectory along the sensor array is generally straight and parallel to the

long axis of the array (X-direction). However, the spot position in the Y direction changes with target distance, for two reasons: 1. Once the light leaves the fast mirror, it travels straight to the target, in the plane that was occupied by L-FOV at that moment. However, while the light travels to the target and back, the S-FOV continues to rotate, and by the time the return pulse reaches the sensor, it has already turned by some angle φ.sub.d; 2. There is some parallax h between the origin points of S-FOV and L-FOV. It is generally small, but if the target is relatively close, it leads to a non-negligible angular deviation of the target from S-FOV plane, and respectively, to the linear displacement of the spot on a pixel in y-direction.

(148) These two effects are illustrated on FIGS. **9**E and **9**F respectively, showing the top view (z-y plane) of the LIDAR system. Dotted lines show the lens and sensor orientation at the Time of Departure, i.e. the moment when the fast mirror had the same elevation as the depicted pixel. It does not matter exactly which pixel and what elevation those are. φ.sub.d can be expressed as:

(149) $_d = {}_2 * t_d = {}_2 * \frac{2*d}{c}$, (9) where d is the distance to the target from which the pulse returned, and t.sub.d is the round-trip ToF to that distance.

Respectively, the Y-position of the spot on the pixel (FIG. **9**E), would shift by:

(150) $_d = FL * {}_d = FL * {}_2 * \frac{2*d}{c}$ (10)

While the magnitude of y.sub.d on FIG. **9**E is exaggerated with respect to the y-size of the pixel, it is still important to center the image of the laser spot on the pixel when the target is far away, and the return signal is the weakest. Therefore, the angle of convergence δ between S-FOV and L-FOV is chosen in such a way, that the light bouncing from the most distant target would land in the center of the pixel, taken into the account the turn of the S-FOV during the round-trip ToF:

(151) $= {}_d + \frac{h}{d} = {}_2 * \frac{2*d}{c} + \frac{h}{d}$ (11)

Numerically, these relationships are illustrated in the Slow Stage sheet of <Sensor Processing Time_1.xlsx>.

(152) FIG. **9**F shows this convergence angle, as well as the effect of the parallax: when the target is relatively near, the image of the laser spot is pushed toward the edge of the sensor and beyond, so at some point it falls off the sensor completely, limiting the short-end range of this type of LIDAR.

(153) Additionally, when the target is sufficiently close, the image of the laser spot on the sensor is getting defocused, and therefore growing larger, since the sensor lens is focused at or near infinity. FIGS. **9**G, **9**H, **9**J, and **9**K qualitatively illustrate the laser spot on the sensor at different target distances.

(154) There is only one target distance at which the laser spot is centered on the sensor and the light collection is maximized (FIG. **9**J). At any other distance, the spot tends to shift to the right due to the two effects described above, however, this shift is relatively small except for very short distances. So, for distances shorter than maximum, the reduction in light collection is compensated by generally stronger signals, while for distances beyond maximum, the signal would be insufficient anyway.

(155) LIDAR Embodiments

(156) FIG. **10** is a perspective view of one embodiment of an optical system of a LIDAR system of the present invention. A laser **103** projects an unscanned beam **103***a* towards a scanning mirror **104** that oscillates about the x-axis, which in turn reflects a scanned beam **104***a* towards mirror **105** which in turn reflects the scanned laser beam **108** forming a laser scanned line **107** on the target **106**. Mirror **105** may be a fold mirror, to make the optical path longer than the size of the optical system. The sensor's field of view with respect to the portion of the target **106** to be imaged is defined by rays **109**, which image is received through lens **102** onto a linear sensor array **101**. (Note—the lens **102** may be an imaging objective or imaging lens, and may be a single lens or a series of such lenses that focuses the spot on the sensors of the linear array).

(157) In FIG. **10** the laser scanned line **107** is shown being substantially centered on the sensor

field of view on target **106**. But this arrangement poses problems for LIDAR systems, particularly for one that may be utilized in a moving vehicle where the distance of objects in the vehicle path often vary significantly (e.g., another car suddenly cutting cross-wise across its path/traffic lane), as described above in the LIDAR Theory of Operation section, and which may occur almost instantaneously. One serious problem is that such an event could suddenly cause so much reflected light to be received by the sensor array that the intensity causes damage to the sensors.

(158) For a LIDAR system, it is disadvantageous to have the optical reception path and the optical transmission path for the laser beam of the system be substantially identical (e.g., being co-axial), as objects in the transmission path (e.g., the mirror **105**) may also block some portion of the light in the reception path (e.g., blocking some of rays **109**—see FIG. **10**). Plus the LIDAR system needs to "see" (i.e., image and determine range) from between one meter to about 200-250 meters, and to also reduce or even eliminate parallax.

(159) The LIDAR system embodiments disclosed hereinafter are particularly configured to address these and other issues.

(160) FIG. **10**A shows the optical system of FIG. **10**, and FIG. **10**B is a two-dimensional image of the optical system of FIG. **10**A viewed along the Z axis perspective; however, the arrangement in each of FIG. **10**A and FIG. **10**B show the scanned laser beam **108** being scanned parallel to the x/z plane with the center of the laser scanned line **107** being somewhat offset from the sensor field of view on target **106**. In practice, the scanned laser beam **108** diverges in the direction of the y-axis and z axis (which divergence is shown in FIG. **10**B), so as the distance to the target increases, the scanned laser beam increases in size and may partially cover (i.e., may bleed into) the sensor field of view on target **106** when the target is at a sufficient distance away. However, the portion of the scanned laser beam **108** that does not fall on the sensor field of view on target **106** provides no benefit to the LIDAR system. When the distance to the target is reduced, and the amount of overlap is reduced such that none of the scanned laser beam **108** falls on the sensor field of view on target **106**, the LIDAR system is essentially blind to objects because of the short distances.

(161) As described, the offset shown in FIG. **10**B between the sensor field of view on target **106** and the laser scanned line **107** may be problematic (depending upon the distance of the object from the LIDAR system), and FIG. **10**C to FIG. **10**J are various embodiments that address this issue, irrespective of the object-to-LIDAR system distance.

(162) In FIG. **10**C, the fold mirror **105** is moved to a position in front of the lens **102** and is appropriately angled so that the center of the scanned laser beam **108** remains parallel to sensor field of view shown by rays **109** (thereby eliminating any parallax issues), however, the distance between the two parallel beams is smaller than that shown in FIG. **10**B. Accordingly, given the position of the fold mirror **105** and the described beam divergence, some portion of the diverging scanned laser beam **108** falls on the sensor field of view with respect to target **106**, and the mirror does not block reflected light in the reception path. Note that the origin point of the light from the mirror **105** is not on the center (i.e., the normal vector **102**N) of the lens **102**).

(163) FIG. **10**D is a further embodiment of the present invention wherein the fold mirror **105** is angled so that the scanned laser beam **108** converges with the sensor field of view rays **109** at some target distance from lens **102**. This distance can be, for example, the maximum working range for the product, for example 200 meters. This would maximize the amount of the scanned laser beam **108** which falls on the sensor field of view (rays **109**) at the systems maximum distance. However, this approach is disadvantaged in that the fold mirror **105** may block some of the return signal to the lens **102**, and when the distance to the target is reduced, the misalignment increases.

(164) FIG. **10**E is a further embodiment of the present invention wherein the fold mirror **105** is positioned well outside the lens **102** field of view and is aligned so the scanned laser beam **108** projects parallel to the normal vector **102**N of lens **102**. The linear sensor array **101** is also positioned offset relative to the normal vector **102**N of lens **102**, and as a result of this offset, the sensor field of view (rays **109**) is offset in the opposite direction relative to the normal vector of

lens **102**N. The scanned laser beam **108** converges with the sensor field of view (rays **109**) at some target distance from lens **102**. This distance can be, for example, the maximum working range for the product, for example 200 meters. This would maximize the amount of the scanned laser beam **108** which falls on the sensor field of view rays **109** at the systems maximum distance. This approach is advantaged as compared with the embodiment in FIG. **10**C in that the fold mirror **105** does not partially block the lens **102**. However, this embodiment is still disadvantaged in that when the distance is to the target is reduced, the misalignment increases. Note that a combination of the arrangements of FIG. **10**D and FIG. **10**E could be utilized, wherein the mirror is generally positioned as shown in FIG. **10**E to be just beyond the distal edge of the lens **102**, but is angled as shown in FIG. **10**D to directly light directly at the target and/or to bleed onto the target.
(165) FIG. **10**F is a further embodiment of the present invention, which does build in some parallax. Elements **101**, **102**, **104***a*, **105**, **106**, **108**, and **109** are all the same as described previously with reference to FIG. **10**B. In FIG. **10**F, an additional optical element **110**, such as a wedge, or negative cylindrical lens, or negative lens with variable cylindrical power, is positioned towards the outside edge of the lens **102** in the direction of the target. (Note that the wedge is shown with a planar surface in FIG. **10**F and may instead be a curved surface). This addition optical element **110** creates an auxiliary field of view **106***a*, scanned light from which would be reflected onto the linear sensor array **101** from nearby targets. This embodiment is advantaged in that when a target is far away from the lens **102**, the light reflected from the sensor field of view **106** dominates and very little return is received along the auxiliary field of view **106***a*. However, when the distance to the target is reduced, the signal along the auxiliary field of view **106***a* increases, and when the distance gets very short, only the signal along the auxiliary field of view **106***a* is reflected on the linear sensor array. In this embodiment, appropriate signal is received by linear sensor array **101** over the entire working range of the device. As an example, the additional optical element **110** can be a negative lens, with curvature in the direction perpendicular to the scan direction. The additional optical element **110** can have a width of 7.5 mm and cover ¼ of the main lens **102** (note that as illustrated in FIG. **10**F and FIG. **10**G, the additional optical element **110** only is shown to cover about one-eighth of the lens **102**). The main lens **102** can be focused at 30 m, roughly its hyperfocal distance. The field of view on the target through the additional optical element **110** is offset from the main lens **102** optical axis by 48 mm and tilted towards the lens by 0.0008 mrad. Note that, as an example, if the center of mirror **105** shown in FIG. **10**F is offset two inches away from the rays **109** (i.e., the center of mirror **102**), then when the target is at a distance of about 200 meters, the laser spot size from the diverging beam may be about 15 inches, Such large laser spot (e.g., a 5 inch spot, a 10 inch spot, a 15 inch spot, etc. will all fall within the sensor FOV). This arrangement uses the disadvantage of the parallax to the benefit of the system, by also allowing the arrangement to take in less light from closer objects.
(166) FIG. **10**G shows the lens **102** and the additional optical element **110** as viewed from the perspective of the x-axis. As a further embodiment of the present invention, the transparency of the additional optical element **110** can be reduced, thereby reducing the collection efficiency at close range. This approach can prevent excessive signals from being received at the linear sensor array **101** when the target is extremely close and therefore the reflected signal is very strong.
(167) Referring to FIG. **10**H, the linear sensor array **101** is positioned such that light reflected from the sensor field of view on target **106** is focused by lens **102** upon the linear sensor array. The light of the laser scan line that is reflected off of target **107** thereafter passes through lens **102**, and is defocused at the linear sensor array **101** but is focused at location **112***a*, which is non-overlapping with the sensor **101**. However, as long as the sensor **101** at least partially extends into the defocused light collected from the target **106**, it would be able to register the reflected pulse, as the total amount of light collected from a nearby target would be much greater.
(168) FIG. **10**J further illustrates an embodiment where the sensor **101** is both extended and tilted (angled) in the direction of the focused image **112***a*, away from the nearby target **106**. In this

embodiment, the light from the target at any range reaches the sensor in a nearly-focused state. Additionally, due to more light being collected from nearby targets, the sensitivity of the sensor may be arranged to gradually drop toward its tilted back side. This arrangement simultaneously corrects for offset and the focusing. (A problem with a fixed focused lens is that as the object(s) get closer and closer, not only is the alignment lost, but so too is the focusing).

(169) FIG. **10**K is an alternative embodiment of FIG. **10**G wherein the additional optical element **110** has variable transparency (shown by lines **110***a*). For example, the transparency of the additional optical element **110** can be more transparent in the direction of the center of the lens **102**, and becomes less transparent as the distance from the center of the lens **102** increased.

(170) In a scanned LIDAR system, care must be taken to make certain that the shape of the projected scan line on the target **107** matches (i.e., is properly aligned with) the sensor FOV on the target. FIG. **11**Ai to FIG. **11**E are various embodiments for insuring proper alignment.

(171) FIGS. **11**Ai-**11**Aiii are a series of orthogonal illustrations showing the unscanned laser beam **103***a*, the scanning mirror **104**, the laser scan line on target **107**, and the scanned laser beam **108** from FIG. **10**Ai. In this example, the unscanned laser beam **103***a* falls solely within the Z-Y plane (has no X axis component) and hits the mirror **104** at a position along the axis of rotation. This arrangement results in the laser scan line on the target **107** being linear and therefore matches the linear sensor FOV **109** from FIG. **10**A, which is desired for the LIDAR system. However, a potential drawback of this type of arrangement is that the scanned laser beam **108** must never be perpendicular to the mirror, to avoid reflecting directly back into the laser **103**, therefore the total angle of the scanned laser beams **108** must be substantially reduced to avoid reflecting back into the laser. To expand the angle of the scanned laser beam and avoid this problem, the components may be arranged to always project at an angle to the mirror's normal (i.e., by not projecting perpendicular to the pivot axis), as discussed in the following embodiments (e.g., compare FIG. **11**Aii and FIG. **11**Bii).

(172) FIGS. **11**Bi-**11**Biii are a series of orthogonal illustrations showing the unscanned laser beam **103***a*, the scanning mirror **104**, the laser scan line on target **107**, and the scanned laser beam **108** from FIG. **10**A. In this example, the origin of the unscanned laser beam **103***a* has a Y-plane offset and hits the mirror **104** at a position along the axis of rotation. In this arrangement the scanned laser beam **108** can be symmetrical about the normal of the mirror **104** without reflecting back into the laser **103**. However, with this arrangement, the laser scan line on the target **107** is non-linear (a curved (“smiley”) shape shown as **107***c*).

(173) FIG. **10**C **11**C is an embodiment that is the same as the embodiment shown in FIG. **11**B, but which includes an additional optical element **102** with variable curvature to straighten the curved smiley shaped scan line **107***c* into the linear scan line **107** for transmission onto target **106**. In this arrangement, the “smiley” is straightened on the outbound path (before reaching the target).

(174) FIG. **11**D illustrates the scanning arrangement of FIG. **11**C wherein the output is a curved smiley shaped scan line **107***c*. In this arrangement an additional optical element **11***a* with variable curvature is placed between the lens **102** and the target and corrects for the curvature of the scan line **107***c* transmitted by lens **102**. Accordingly, the sensor **101** receives a linear reflection of the curved smiley shaped scan line **107**C. In this arrangement, the “smiley” is straightened on the inbound path (before reaching the sensor array).

(175) FIG. **11**E illustrates the scanning arrangement of FIG. **11**B wherein the output is a curved smiley shaped scan line **107***c*. In this arrangement the pixels on the sensor **101***a* are arranged in a curved pattern to match the curvature of the received reflection of the curved smiley shaped scan line **107**C. In this arrangement, the “smiley” is straightened upon reaching the sensor array, by the configuration of the sensors of the array.

(176) In a LIDAR system, it is desirable to move the mirror 80,000 times per second, and a deterrent to such rapid mirror movement is the mass of the mirror itself. So it is desirable to have the mirror be as light as possible, but it is also undesirable to have the mirror flexing while

undergoing such movement. The following embodiments provide improvements to the mirror to better enable the desired mirror speeds.
(177) The Applicant's co-pending patent application Ser. No. 17/000,464, filed Aug. 24, 2020, having the title “Distally Actuated Scanning Mirror,” includes a FIG. **8**B which describes as a mirror structure wherein material was strategically removed on the backside to reduce its weight.
(178) FIG. **12**Ai herein shows such a mirror before processing, where the mirror has been formed out of a silicon on insulator (SOI) wafer, having a handle layer **301***a* and a device layer **301***b*. A photo resist mask **303** is deposited at locations where the handle layer **301***a* is to remain intact. The photo resist mask **303** is not affected by etchants. An etchant **305** is temporally applied and removes a very well defined depth of the handle layer **301***a* except at the positions where photo resist mask **303** is located. The resulting mirror structure is shown in FIG. **12**Aii.
(179) FIG. **12**Bi shows the mirror before processing, where the mirror has been formed out of a single layer silicon wafer **302**. An oxide mask **304**, which etches at a slower rate than silicon, is deposited at locations where the silicon wafer **302** is to remain intact. An etchant **305** is applied and removes the silicon wafer **302** except at the positions where the oxide mask **304** is located. The resulting mirror structure is shown in FIG. **12**Bii. In one embodiment, the etchant **305** can be selected such that it etches the silicon **302** at a rate of seven times faster than the rate it that it etches the oxide mask **304**.
(180) FIG. **12**Ci is an embodiment wherein the mirror shown in FIG. **12**Bii is processed a second time. During this second processes step, an oxide mask **304**, which etches at a slower rate than silicon, is deposited at locations where the silicon wafer **302** is to remain intact. An etchant **305** is applied and removes the silicon wafer **302** except at the positions where the oxide mask **304** is located. The resulting mirror is shown in FIG. **12**Cii. As can be seen, the silicon material removed is at various depths thereby permitting mirror structures that are lighter while maintaining structural integrity. This process can be repeated multiple times to achieve even more variations in the various different depths to which the material is removed.
(181) FIG. **12**D is an image of a scan mirror after an etchant has selectively removed portions thereof, and forms a honeycomb structure (i.e., which may include the depth steps of FIG. **12**Cii), which has most of the material removed at the periphery (i.e., the “wings”), and not from the center section between the hinges, which transmits the torque from the hinges to the outer areas. The wings may progressively get thinner and lighter the more distal from the center, being limited by stress, fatigue, and other design considerations.
(182) FIG. 2 of the Applicant's co-pending patent application Ser. No. 17/000,464 shows a mirror including a mirror block **201** and a hinge **202**.
(183) FIG. **12**E shown herein is an embodiment providing an improvement to the mirrors of the '464 patent application, which mirror includes a mirror active surface **310** and torsional hinges **311**. The torsional hinges **311** connect to a flexure **313** which is mounted to piezoelectric elements **314**. The piezo elements **314** are electrically driven to induce oscillation of the mirror active surface about the axis of the hinges. Curved transition areas **312** are located where the mirror active surface **310** connects to the hinge **311**. The transition area **312** is tapered and provides significantly more strength as compared to a right-angle transition area. This increased strength permits mirror active surface **310** to be oscillated at a higher frequency and/or driven at a wide angle.
(184) FIG. **12**F illustrates another improved embodiment, where torsional hinges **311** taper down toward the mirror active surface **310**. Since hinges themselves may have a considerable amount of inertia being comparable to the mass of the active surface, and oscillate synchronously with the active surface, the inertia of the proximal parts of the hinges contributes to the torque in their distal parts, and therefore concentrates the stress and creates a possible failure point at the distal ends. Tapered hinges alleviate this problem, by restoring a nearly-uniform stress distribution, where the hinges are thicker at the root, and have a curved transition **312** to reduce a stress concentration at the joining to the active mirror surface and to the piezo elements **314**.

(185) During formation of the mirror (i.e., silicon processing), micro imperfections may be formed and may remain in the surface. FIG. **12**G shows a cross section of a hinge **317** which is a high stress silicon MEMS structure. The microcracks **318** that can be present have a detrimental effect on the strength and lifetime of the hinge. FIG. **12**H is an embodiment illustrating an approach to reduce the size of the imperfection, i.e., reduce the size of any microcracks **318**. The structure is oxidized thereby providing an oxidization layer **319** over the structure as well as the microcrack **318**. FIG. **12**J shows the hinge of FIG. **12**H after the oxidization layer is removed. The microcrack is reduced in size and is shown as reduced microcrack **318***a*. This process of FIGS. **12**H and **12**J can be repeated several times to reduce the microcracks down to a negligible size.
(186) The mirror surface is required to reflect the laser light with a high degree of efficiency, and silicon does not transmit the frequency of light that is desirably utilized in the LIDAR system disclosed herein; thus a coating for the mirror is needed, which may be a dielectric coating or gold plating. The dielectric coating may be most advantageous as it may transmit 99.9 percent of the desired wavelength, whereas the gold plating is only about 98 percent reflective, which can produce a detrimental amount of heat when using 10 watts of power to satisfy of design considerations of the LIDAR system. FIG. **12**K shows a cross section of the mirror **310** with a high-reflectivity (e.g., dielectric) coating **320** applied. The coating acts like a Bragg reflector reflecting the specific laser frequency applied. A multilayer dielectric coating can provide reflectivity of 99.8%, however, the thickness of such coating is on the order of 4.5 um thick and has a tendency to bow the entire mirror, by introducing mechanical stress.
(187) FIG. **12**L shows an embodiment wherein a stress-relief sub-layer **321** (for example, chromium) is deposited between the high-reflectivity dielectric coating **320** and the mirror **310**. The stress-relief sub-layer **321** bows the mirror in the opposite direction of the bow caused by the high-reflectivity dielectric coating **320**. The bows cancel each other out and the resulting mirror is flat.
(188) An alternate embodiment for a silicon mirror wherein the laser light is at 1500 nm is to use a gold plating to provide about 98% reflectivity. Silicon does not absorb 1500 nm light and most of the 2% of the laser light that does not get reflected will simply pass through the silicon mirror, thereby limiting any heating effects. In this embodiment, care must be taken to make certain that any photons passing through the silicon mirror do not interfere with other aspects of the design. Light catches can be included in the design to capture any stray photons that are not reflected.
(189) FIGS. **12**M and **12**N, and **12**P and **12**Q schematically illustrate different ways to drive the mirror shown in FIGS. **12**E and **12**F, using piezoelectric elements **314***a* and **314***b*, respectively, which are driven in different directions. (Note, elements **314***c* are piezo benders, and elements **315** are spacers of any solid material).
(190) FIG. **12**R illustrates a laser **103** projecting an unscanned beam **103***a* towards a mirror **104** which scans the beam towards mirrors **322** which are arranged in the arcuate pattern shown to reflect the beam towards a target as the scanned laser beam **108** in turn forms the laser scan line on target **107**. FIG. **12**S illustrates a laser **103** projecting an unscanned beam **103***a* towards a rotating polygon **323** which scans the beam towards mirrors **322** which are arranged to reflect the beam towards a target as the scanned laser beam **108** with in turn forms the straight laser scan line on target **107**. The arrangements shown in FIG. **12**R and FIG. **12**S can be utilized in the prior embodiments to form the laser scan line on target **107**. The arrangements of FIG. **12**R and FIG. **12**S each utilize a mirror with relatively small frequency, but very high scan angle, and it splits the large scan angle into several small sub-angles that subsequently overlap, resulting in a high scan frequency at modest angles, as needed, but using the mirror with lower frequency and higher scan angle.
(191) FIG. **12**T illustrates an arrangement whereby the mirror **104** scans unscanned laser beam **103***a* through telescopic lens elements **330** having power either one or two dimensionally. The selection of the telescopic lens elements **330** permits a tradeoff between the scanned laser spot size and the effective angle of the scan. This arrangement creates a scanning telescope, ending up with a

smaller FOV and a higher optical resolution within the FOV. So greater range is obtained that keeps the same resolution. For example, in a LIDAR system that uses 512 pixels to cover a 30 degree fOV, use of this arrangement may use the same 512 pixels to obtain the same resolution, but would only cover a 10 degree fOV. This arrangement in FIG. **12**T also provides a longer range for the optical system of the LIDAR apparatus (i.e., it trades angle for range).

(192) FIGS. **13**A-**13**H show various embodiments for generating the slow speed scan stage of the present invention, which are alternatives to having 360 degrees of rotation, and also relates to correcting the curved “smiley” scan line discussed hereinabove. In each figure, the laser transmitter **401** is shown which produces the vertical high-speed fan of rays **335** and a receiver **402** is shown with a vertically-oriented sensor array. The laser transmitter **401** and the receiver **402** are optically arranged in each case so that the target field of view illuminated by the vertical high-speed fan of rays **335** corresponds to the field of view of the receiver **402**. The arrangement may have a limited scan angle, being within plus or minus 30 degrees or plus or minus 120 degrees in the forward direction. Also, both the laser beam and the FOV of the receiver may be passing through the same slow mirror.

(193) FIG. **13**A show the transmitter **401** and the receiver **402** mounted to a rotating platform **400**. Electrical signals are transmitted from the components on the rotating platform through slip rings **403***a* and **403***b*. In this case slip ring **403***a* is mounted to the platform and rotates therewith and slip ring **403***b* is fixed. FIG. **13**B shows an embodiment wherein the slip rings of FIG. **13**A are replaced by wireless communications, such as, for example, radio communication, optical communication or magnetic communications. In this embodiment, the wireless transmitter/receivers are shown as **404** and **405**.

(194) FIGS. **13**C-**13**H show embodiments wherein the transmitter **401** and the receiver **402** are rigidly mounted and the motion in the slow speed direction (orthogonal to the vertical high-speed fan of rays **335**) is generated by moving mirrors. In FIG. **13**C, a mirror **406**, which is large enough to receive/accommodate both the vertical high-speed fan of rays **335** and the optical field of view of receiver **402**, is positioned with the axial direction of its face to be perpendicular to the center axis of the vertical high-speed fan of rays **335**. The axis of rotation of the mirror **406** is along the axis or the direction of the scan of the high-speed fan of rays **335**. In this embodiment it should be noted, the two-dimensional field of view is not bowed (no smiley), however, the scanning of the mirror **406** cannot be symmetrical about the plane formed by the high-speed fan of rays **335**, in order to avoid projecting directly back at the transmitter **401** and receiver **402**.

(195) FIG. **13**D shows an embodiment similar to the one shown in FIG. **13**C, but wherein the mirror **406***b* has an axis of rotation that is not perpendicular to the center ray of the high-speed fan of rays **335**. In this case, the mirror can oscillate symmetrically as the scan of the high-speed fan of rays **335** of the mirror **406***b* misses both the transmitter **401** and the receiver **402**. However, this approach is disadvantaged in that the optical arrangement results in the two-dimensional field of view being bowed (has a smiley). According, post processing of the image is required to correct for the bowed field of view.

(196) FIG. **13**E shows a further embodiment that is similar to the one shown in FIG. **13**C, but where the mirror **406***c* has an axis of rotation which is non-parallel to the mirror **406***c* plane (i.e., being at an acute angle thereto). In this embodiment, the mirror **406**C is shown mounted to a cone **409** having a larger diameter on the bottom **409***a* and a smaller diameter **409***b* on the top. The axis of rotation **409***c* of the cone **409** is vertically through the center of the cone **409**. This approach is advantaged in that the scan can be symmetrical about the high-speed fan of rays **335** since the mirror **406***c* is tilted, so that the outgoing rays do not reflect directly back into the transmitter **401** and the receiver **402**. Further, the approach is advantaged in that there is no bowing or smiley due to the optical geometry. While this embodiment shows the mirror **406***c* mounted on a cone **409**, other mechanical embodiments can be utilized to provide that the axis of rotation is non-parallel to the mirror **406***c* plane. For example, the mirror can be mounted on a triangle wedge which in turn

oscillates.
(197) FIG. **13**F shows a further embodiment wherein an additional slow speed mirror **408** rotates about an axis perpendicular to the high-speed fan of rays **335**, and also prevents having the leaser beam being directed back upon itself. The additional slow speed mirror is positioned to receive the high-speed fan of rays **335** and directs it towards mirror **406** which rotates about an axis that is coplanar with the fan of rays **335**. The oscillation speed of the additional low speed mirror **408** may in one embodiment be twice as fast as the oscillation of mirror **406**. This embodiment is advantaged in that there is no bowing (no smiley).
(198) FIG. **13**G shows a further embodiment wherein a gimbled mirror comprises a mirror portion **406***d* which scans about with respect to a first axis **406**Dx that is perpendicular to the high-speed fan of rays **335**. An outer frame portion **406***e* scans the mirror portion about a second axis **405**Ex which is coplanar with respect to the fan of rays **335**. The oscillation speed of the mirror portion **406***d* may in one embodiment be twice as fast as the oscillation of outer frame portion **406***e*. This embodiment is advantaged in that there is no bowing (no smiley).
(199) FIG. **13**H shows a further embodiment wherein multiple segmented mirrors **406**F are vertically positioned wherein each scans about an axis oriented in line with the fan of rays **335**. One of the segmented mirrors **406**F, such as for example the most centered segment mirror is arranged to receive the high-speed fan of rays **335** and directs it towards the target. The remainder of the segmented mirror **406***f* optically transmit the return signals from the field of view to the receiver **402**. This approach is advantaged in that there is no bowing (no smiley). Furthermore, each section of the segmented mirror is much smaller than a single mirror, and therefore has less mass that needs to be rotated.
(200) FIGS. **14**A through **14**Dii show various embodiments of the sensor arrays that may be used in the LIDAR system disclosed herein.
(201) The sensor arrangement of FIG. **14**A is configured to address the an issue that occurs during the scanning of the laser beam as discussed herein, whereby the scanner moves at different speeds at the center of the scan than at the end of the scan (i.e., the scanner must slow down at it reaches the extent of the oscillation in each direction, and may be at a maximum speed in the center). Thus, there is more signal obtained when in the center, and is not perfectly uniform. The sensor arrangement of FIG. **14**A compensates for those differences in scan speed for the outbound laser light. In FIG. **14**A, the individual pixels **101***b* of sensor array **101** vary in size and shape, but have each center positioned in a line. In this embodiment the middle pixels are approximately square and each successive pixel in moving outwardly from the middle is shaped to be longer and narrower. The actual laser spot when being mechanically scanned can vary in shape and scan speed, as the scanning progresses out from the center point of the scan, the middle pixels are the widest for when the scanned beams moves the fastest, and the width of each successive pixel gets narrow as the slowing laser beam increasingly spends more time on the narrower sensor. The variations of the pixels **101***b* can be set to compensate for such changes in the scan spot size and speed, thereby providing uniform detection outputs. As a further embodiment, the spacing between the individual pixels **101***b* can also be varied to match the actual scanning patterns and scan speed.
(202) FIG. **14**B describes a sensor array **101** wherein the individual pixel sensors **101***b* are arranged in a bowed (smiley) arrangement. As previously described, there are embodiments wherein the scanned laser line **107** is bowed. By bowing the individual sensor **101***b* positions on the sensor array **101** to match the bowed laser line **107**, the LIDAR system does not need to perform any post image processing to correct for such bowing of the laser line **107**.
(203) As previously described, there are some situations wherein there is parallax between the scanned laser and the optical field of view. FIG. **14**C shows an embodiment for dealing with this parallax effect/error. In this embodiment, the image sensor **101** is formed by a column of smaller linear sensors **101***b* and a second column of larger sensors **101***c*. The smaller linear sensors **101***b* can be positioned so that the field of view is in focus at a far field positions, for example 25-200

meter from the sensor. The larger sensors **101***c* can be positioned so that the field of view is in focus at a near field positions, for example 0-25 meter from the sensor. As a further embodiment, the small sensors **101***b* can be a different type of sensors as compared to the larger sensors. For example, it is often required that the smaller sensors **101***b* detecting the far field object are more sensitive. According, the smaller sensors **101***b* could be either APDs, SPADs or other high sensitive sensors. The larger sensors **101***c* for detecting the objects in the near field generally receive larger reflection signals, due to the objects being relatively close, and therefore do not need to be as sensitive. Accordingly, the larger sensors **101***c* can be conventional photodiodes.
(204) A challenge when utilizing high powered scanning lasers with very sensitive image sensors is that the sensors can be damaged if a strong reflection from the scanned laser beam is received or if another laser system shines its laser light directly into the image sensor pixels. FIG. **14**Di is another embodiment of the image sensor **101** that includes guard rail sensors **101***d* that surround the individual sensors **101***b* of the array. The guard rail sensors **101***d* can be configured to have low sensitivity and a high immunity to damage when laser light is received. Generally, in order for a scanned laser to reach the highly sensitive individual sensors **101***b*, the laser light would first pass over the guard rails sensors **101***d*. Electronics are also provided that detect the laser light falling on the guard rail sensors **101***d* that exceeds a threshold intensity level, and temporarily shuts down the individual pixels **101***b*, thereby avoiding damage as the laser moves over the highly sensitive individual pixels **101***b*. The guard rail sensors **101***d* can be formed by standard photodiodes. The spacing between the guard rail sensors should be less than the extent of the laser spot produced by an offending LIDAR system or other system. Assuming that the spot is well-focused, which is unlikely, but not impossible, the guard rail sensors would need to be spaced apart no more than about tens of um.
(205) As a further embodiment (see FIG. **14**Dii), the plurality of guard rail sensors **101***d* that number greater than the sensors of the linear array in FIG. **14**Di can be replaced by four photodiodes that surround the individual pixel sensors **101***b*, two long ones for the lengthwise direction and two short ones to cover the width of the sensor **101**. Still further, the guard rail can be formed by a single sensor shaped to surround the individual sensors **101***b.*
(206) FIG. **14**E illustrates each that each individual APD pixel **101***b* shown in FIG. **14**Di and FIG. **14**Dii may be connected to a transimpedance amplifier (TIA) **501** that is connected in turn to a threshold comparator **502** which in turn connects to a time stamp **503** circuit. When a rising edge signal (i.e., detected laser light) is detected as being above a preset threshold by the threshold comparator **502**, the time stamp **503** circuit records the rising edge time. When a falling edge signal is detected below a preset threshold by the threshold comparator **502**, the time stamp **503** circuit records the falling edge time. The rising edge time and the falling edge times are communicated to a serializer which provides a serial output including the rising edge time and the falling edge time for each APD pixel **101***b.*
(207) The embodiment shown in FIG. **14**F is similar to the one shown in FIG. **14**E except that a second threshold comparator **502***a* and a second time stamp **503***a* are added to the processing channel for each APD pixel **101***b*. The threshold setting for the second threshold comparator **502***a* is different than the threshold setting for threshold comparator **502**. When a rising edge signal is detected above a preset threshold by the threshold comparator **502***a*, the time stamp **503***a* circuit records the rising edge time. When a falling edge signal is detected below a preset threshold by the threshold comparator **502***a*, the time stamp **503***a* circuit records the falling edge time. The rising edge time and the falling edge times from both time stamp **503** and second time stamp **503***a* are communicated to a serializer which provides a serial output including the rising edge times and the falling edge times associated with threshold comparators **502** and **502***a*. While this embodiment is shown with two threshold comparators and two time stamps for each pixel channel, it is possible to add still further threshold comparators and time stamps to each pixel channel.
(208) FIG. **14**G is an embodiment wherein the amplified signals from pixels **101***b* are not just

compared to one or more preset threshold values, but are also digitized as sequences of samples using analog-to-digital converters (ADC) **503**. The threshold comparators **502** are still present, however their role is limited to detecting the beginning of a pulse from a pixel and assigning one of the available ADC channels to that pixel via cross-bar **506**. The ADC channel remains assigned until the signal drops below the threshold on the comparator **502**, upon which the ADC channel is freed up and become available to other pixels. Since return pulses detected by pixels **101***b* are generally short and infrequent, the number of ADC channels may be substantially smaller than the number of pixels. The serializer **504** provides a serial output where each pixel event is represented as: a pixel index, number of samples, time stamp of the first sample and sample values.
(209) Fiber laser outputs are typically circular and non-astigmatic, and therefore require relatively simple collimation arrangements.
(210) FIG. **15**A illustrates light from a fiber laser **103***b* passing through an aspheric lens **601** and then being scanned by mirror **104**.
(211) Conversely, diode lasers typically produce highly elliptical beams, and therefore additional circularizing optics are required to shape such a beam to fill a circular mirror. For example, FIGS. **15**Bi and **15**Bii illustrates light emitted from a diode laser **103***c* passing through cylindrical lens **602** and then through an aspheric lens **601** and then being scanned by mirror **104**.
(212) The output beam of a Master Oscillator Power Amplifier (MOPA) laser is typically not only elliptical, but also astigmatic, therefore even more complex optics is needed for collimation.
(213) FIG. **15**Ci and FIG. **15**Cii illustrate light from a MOPA laser **103***d* passing through a cylindrical lens **602***a* for fast direction MOPA collimation and then passing through cylindrical lens **602***b* for slow direction MOPA collimation, and then being scanned by mirror **104**.
(214) FIG. **15**Di and FIG. **15**Dii illustrate a more detailed embodiment for focusing a MOPA laser with one aspheric and one cylindrical lens.
(215) The embodiment shown in FIG. **15**Ei and FIG. **15**Eii is similar to the one shown in FIG. **15**Ci and FIG. **15**Cii, but with the addition of a low-reflectance signal sampling mirror **603** which diverts some of the light beam coming out from the cylindrical lens **602***a*. The diverted light beam is imaged by a dynamic imaging optical sensor **604** for astigmatism control. Accordingly, the system can be adjusted based upon the detected astigmatism so as to appropriately shape the beam to be scanned by mirror **104**. A possible adjustment method is through the current injected into the Amplifier section of the MOPA laser. Changing the current utilized typically changes the astigmatism, so that it may be brought back to a more acceptable amount. Alternatively, the astigmatism could be adjusted mechanically/optically by moving one or both of the lenses, rather than adjusting it electronically.
(216) FIG. **16**A is a control system embodiment that may be used for accurate control of the motion of the high-speed scanning mirror **104**, which includes, in addition to the main laser, a pickoff laser **701**, which has much lower power and emits visible light rather than IR light, which visible light is reflected off the same mirror **104** and forms a scanned laser beam **702** which is detected by two end pickoff sensors **704***a* and a center pickoff sensor **704***b*. The signals from the pickoff sensors can be utilized to determine the angular position of the mirror **104**. The pickoff laser **701** is positioned in a way that makes its fan of scanned rays out of the plane with the fan of the rays **335** from the main scanning laser **103**. FIG. **16**B shows the mirror scan profile **705** and the pulses from the end pickoff sensors **706***a* and the pulses for the center pickoff sensors **706***b*. FIG. **16**C is the same a FIG. **16**B but includes a digital representation of the laser on interval **707** and the laser off interval **708**.
(217) FIG. **16**D is a piezo drive comprising a control circuit **713** connected to a resonant power driver **712** which connects through inductor **711** to the piezoelectric element **314** which in turn drives the mirror. The same circuit can drive multiple piezoelectric elements connected in series or in parallel.
(218) Some types of piezoelectric elements that may be used to drive movement of the scanning mirrors disclosed herein, for example, multi-layer stacks working in d33 mode, require biased

electrical drive signals, so that the applied voltage is never negative with respect to the polarity of the piezoelectric element.

(219) FIG. **16**E is an example of a biased drive circuit wherein the resonant power driver **712** connects through inductor **711** to two diodes **714**, having opposite polarities, which connect in turn to two capacitors **71** which in turn drive the piezoelectric elements **314** of the mirror.

(220) FIG. **16**F is an arrangement wherein different piezo elements **314** are driven at different phase. In this case, each of four piezo elements **314** are connected through an inductor **711** to the output of a resonant power driver **712**. Since different pixels are illuminated by reflected light that was emitted at different positions of the scanning mirror, the sensor electronics responsible for ToF measurement must be synchronized with the mirror movement.

(221) FIG. **16**G is an arrangement wherein the sensor timing is driven off the end pickoff pulse **706***a*. Since the actual time that elapses from the moment when the mirror is pointing toward the pickoff sensor, to the moment it is pointing toward the target point visible to the given pixel indexed by pixel position **724**, the sensor electronics responsible for ToF measurement must also be aware of the current frequency of the mirror **725**, which may change somewhat during operation Mirror current frequency **725** and pixel position **724** are provided as input to the scan profile LUT **722**, resulting in the accurate measurement of the ToF.

(222) FIG. **16**H is an arrangement where a PLL **723** is utilized to smooth the pickoff pulse **706** in the event that any jitter occurred in the pickoff pulse **706**.

(223) As previously described, the high speed scanned column has a slow stage component for providing the horizontal direction of scanning. FIGS. **17**A-**17**E illustrate examples wherein the horizontal slow stage of scanning is implemented by rotating the high-speed linear scanned column. In FIG. **17**A an embodiment is illustrated wherein the range of the device is selectively altered during various stages of the image. In this example **801***a* represents the forward scans in the 0-180 degrees of a 360-degree scanner, and **801***b* represents the rear scans in the 180-360 degrees of the 360-degree scanner. The laser power output when forward scanning **801***a* can be increased thereby increasing the laser energy falling on the target, and thereby effectively increasing the range of the LIDAR system. This might be desirable, for example, in a car, where the system wants maximum range forward of the car. The laser power output when rear scanning **801***b* can be decreased thereby decreasing the laser energy falling on the target, and thereby effectively reducing the range of the LIDAR system. This might be desirable, for example, in a car, where the information to the rear of the car is less critical and a lesser range is required. Reducing the laser power during portions of the scan has the effect of reducing the overall power consumed by the LIDAR system.

(224) In FIG. **17**B an embodiment is illustrated wherein the effective vertical FOV of the device is selectively altered during various stages of the image, by altering the time during which the laser is on. In this example **802***a* represents the forward scans in the 0-180 degrees of a 360-degree scan, and **802***b* represents the rear scans in the 180-360 degrees of the 360-degree scan. When forward scanning that produces forward scans **802***a*, the laser is energized for a longer time, thereby maximizing the number of pixels in each column scanned, and thereby effectively maximizing the resolution of the LIDAR system. This might be desirable, for example, in a car, where the system wants maximum resolution in the direction forward of the car (i.e., the direction of travel). When rear scanning that produces scans **802***b*, the laser is on for a smaller portion of the vertical high-speed scans thereby decreasing the number of illuminated pixels and respectively, the vertical resolution of the system. This might be desirable, for example, in a car, where the information to the rear of the car is less critical and a high resolution is not required. Turning off the laser power for shorter periods during vertical high speed has the effect of reducing the overall power consumed by the LIDAR system.

(225) FIG. **17**C is an embodiment where the laser power applied during various high-speed vertical scans is varied. For example, every 4th vertical scan can be high powered and the remaining

vertical scans can be at low power. In this manner the entire image field has interlaced long and short range scans. Turning down the laser power during some of the vertical high speed scans has the effect of reducing the overall power consumed by the LIDAR system.
(226) In FIG. **17**D an embodiment is illustrated wherein the vertical scan angle of the device is selectively altered during various stages of the image. In this example **803***a* represents the forward scans in the 0-180 degrees of a 360-degree scan, and **803***b* represents the rear scans in the 180-360 degrees of the 360-degree scan. When forward scanning results in scans **803***a*, the maximum angle of scan in the vertical direction is enable. When rear scanning **803***b*, the scan angel of the vertical high-speed scans is reduced.
(227) The embodiments shown with respect to **17**A-**17**D were illustrated with 360-degree horizontal devices. The concepts presented herein could equally apply to other horizontal fields of view, such a 120-degree or 180-degree fOV LIDAR system. Further, while just two zones were shown (front facing and rear facing), this invention is not so limited. Multiple zones could be implemented, each have a varied field of view. Each zone can have its own range and/or resolution. Furthermore, each of the inventions shown in FIGS. **17**A-**17**D can be combined in any combination into a single LIDAR system. For example, the forward-facing view can have both increased laser power and the maximum resolution, while the rear facing view can have both the reduced laser power and the reduced resolution.
(228) FIG. **17**E is an embodiment wherein two LIDAR systems are incorporated into one LIDAR package. A narrow vertical angle device with long range **804***b* can be combined with a wide vertical angle short range device **804***a*.
(229) FIG. **17**F shows a car that uses mounting structure **807** to support a LIDAR system **808** as disclosed herein, which LIDAR system may, for example, be configured to have a forward facing view and forward scans, and a rear facing view and rear scans.

Claims

1. An optical system for a light detection and ranging (LIDAR) system comprising: a laser configured to emit a beam of light at a wavelength; an optical transmission system configured to shape the beam of light emitted by said laser, said optical transmission system comprising: a first mirror, said first mirror configured to pivot about an axis to scan said beam of light; and a fold mirror, said fold mirror positioned to reflect said scanned beam of light along a fan of transmission light paths toward a target; an optical reception system configured to collect said laser light reflected from the target along a fan of reception light paths, said optical reception system comprising: a lens; and a sensor array, said sensor array having a field of view (FOV) on at least a portion of the target a wedge-shaped optical element configured to create an auxiliary field of view; wherein said wedge-shaped optical element is positioned proximate to an outside edge of said lens; and wherein said fold mirror is positioned proximate to, but beyond said outside edge of said lens.
2. The optical system according to claim 1, wherein said wedge-shaped optical element comprises a width configured to cover one-quarter of said lens.
3. The optical system according to claim 2, wherein said wedge-shaped optical element comprises a variable transparency, having increasing transparency with increasing distance from a center of said first lens.
4. The optical system according to claim 3, wherein said scanner comprises a mirror configured to pivot about an axis to scan said beam of light toward said fold mirror.
5. The optical system according to claim 4, wherein said fold mirror comprises a plurality of mirrors arranged in an arcuate pattern to reflect said scanned beam of light toward the target.
6. The optical system according to claim 5, wherein said scanner comprises a polygonal reflector configured to rotate about an axis to sequentially scan said beam of light toward each of said plurality of mirrors arranged in said arcuate pattern.

7. The optical system according to claim 1, wherein said optical transmission system further comprises: a pair of telescopic lens elements.

8. The optical system according to claim 1, wherein said sensor array comprises: a plurality of sensors each having a unique length and width; and wherein a center of each of said plurality of sensors are arranged in a line to form a linear sensor array.

9. The optical system according to claim 1, wherein said sensor array comprises: a plurality of sensors each having the same length and width; and wherein a lengthwise direction of each of said plurality of sensors are parallel, and a center of each of said plurality of sensors are positioned to form an arcuate shape.